

7-Zip Security Assessment Report

**Threat Modeling, Vulnerability Analysis, and Secure
Software Engineering Review**

Tharusha Dilshan

Cyber Security Undergraduate

Sri Lanka Institute of Information Technology (SLIIT)

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1 Executive Summary

This report presents a detailed security assessment of 7-Zip, a widely used open-source file compression and extraction utility developed by Igor Pavlov. The main purpose of this study is to understand how real-world software systems handle security and to identify common weaknesses that can emerge in production-level code.

The assignment focuses on three major areas: Domain and Historical Analysis, Design Analysis, and Code Inspection. First, the domain analysis explores how 7-Zip functions, the type of data it handles, and what assets require protection, such as compressed archives, encryption keys, and temporary files. It also reviews previously reported vulnerabilities and how they were addressed by the development community.

The design analysis evaluates the internal architecture of 7-Zip, including its compression engine, encryption module, and user interface, while identifying possible threat surfaces through a structured threat model. Finally, the code inspection stage examines selected source files to locate security-related coding issues such as memory handling errors, buffer overflows, and unsafe function calls.

Overall, this report aims to apply secure software engineering concepts to a real, open-source product and provide practical insights into how tools like 7-Zip can be made more robust, resilient, and trustworthy for everyday users.

2 Introduction

2.1 Assignment Objective

This assignment focuses on carrying out a security review of 7-Zip, an open-source file compression and extraction software that is widely used in day-to-day computing. The main goal is to understand how real-world software systems deal with security challenges and how different types of weaknesses can appear during development. By studying 7-Zip, I aim to apply the theoretical concepts learned in class to a real application and see how issues such as poor coding practices, architectural flaws, or weak security mechanisms can lead to vulnerabilities. The work involves reviewing its structure, examining past security incidents, and going through parts of its source code to identify potential risks. Through this process, I hope to gain practical knowledge about secure software development, learn how vulnerabilities can be detected before release, and understand how such findings can improve the overall reliability of a product.

2.2 Scope and Methodology

The study is based on analyzing the public documentation, design, and source code of 7-Zip. It was selected because it is free, open source, and used by millions of people for compressing and extracting files in various formats like ZIP, RAR, and 7z. Since 7-Zip has existed for many years, there are also several publicly reported vulnerabilities and bug-fix histories, which make it suitable for a case study.

The work mainly focuses on three areas:

- **Domain and Historical Analysis** – This part discusses what 7-Zip does, who uses it, and what key assets it handles, such as compressed archives, encryption keys, and temporary files. It also looks at some of the major security incidents in its past and what lessons can be learned from them.
- **Design Analysis** – This part studies 7-Zip's architecture, including components like the compression engine, encryption module, and graphical interface. It also develops a threat model to highlight areas where attackers might exploit weaknesses or cross trust boundaries.
- **Code Inspection** – This step reviews selected pieces of 7-Zip's source code to identify insecure coding patterns, focusing mainly on input validation, memory management,

and encryption logic. The goal is to highlight issues such as buffer overflows or unsafe API calls that could lead to real-world exploitation.

Through these steps, this report provides an in-depth evaluation of 7-Zip's security posture and suggests possible improvements aligned with secure software engineering principles.

2.3 Tools and Frameworks Used

To conduct the security evaluation of 7-Zip, a combination of industry-standard and open-source tools was employed. Each tool played a distinct role in identifying design-level weaknesses, source-level vulnerabilities, and code-quality issues across different stages of the study.

Threat Modeling: The Microsoft Threat Modeling Tool (TMT) was utilized to design the data-flow and trust-boundary diagrams for 7-Zip. It systematically identified potential attack surfaces, untrusted data paths, and privilege boundaries in the application architecture. This enabled early visualization of threats before code-level analysis.

Static Code Analysis:

A multi-tool approach was adopted to improve accuracy and reduce false positives:

- **Snyk Code** – Performed semantic analysis to detect memory management issues such as use-after-free, double free, and memory leaks (CWE-401, 415, 416).
- **SonarQube** – Scanned the entire codebase for structural defects, maintainability issues, and security bugs, producing a prioritized severity matrix.
- **Semgrep.dev** – Applied lightweight, rule-based detection aligned with CERT C and CWE standards, identifying unsafe file-permission operations and string mismatches.

Version Control and Code Provenance:

Commit histories from Git Log and SourceForge repositories were examined to trace vulnerability fixes and understand the evolution of security practices within 7-Zip's development lifecycle.

Together, these frameworks provided a complete security-engineering workflow—from architectural threat identification (Microsoft TMT) to automated code inspection (Snyk, SonarQube, Semgrep) and verification of vulnerability patches through source control.

3 Product Selection

3.1 Criteria for Software Selection

Selecting suitable software for a security analysis is one of the most crucial steps in this assignment, as it directly influences the depth and relevance of the results. In this study, the goal was to identify an open-source application that is widely used, technically complex, and has a history of real-world security concerns. The chosen software also needed to provide publicly available source code and developer documentation that could support analysis of vulnerabilities and code-level security practices.

First, the software had to be open-source and publicly accessible so that its source code could be freely examined. This allows not only an understanding of how the system functions internally but also enables tracing past vulnerabilities and reviewing how the development team handled previous security issues. Having the ability to inspect actual implementation files makes it easier to perform an in-depth security review and identify design flaws that could otherwise remain hidden.

Secondly, the software needed to be actively used in real-world environments rather than being purely experimental or research based. Applications that are part of daily user activity tend to face continuous exposure to different attack vectors, which provides a more realistic view of how security mechanisms hold up under active use. 7-Zip meets this requirement since it is a trusted tool for compressing and extracting files on millions of computers worldwide and is used in personal, business, and industrial environments.

Third, the selected software had to carry meaningful security relevance within its domain. 7-Zip handles sensitive data through file compression, encryption, and decompression processes. Improper handling of input files or cryptographic routines could expose users to threats such as arbitrary code execution, data leakage, or privilege escalation. Its encryption features and file parsing functions therefore make it an excellent subject for identifying vulnerabilities in secure software development.

Fourth, the system had to be large and complex enough for analysis—ideally consisting of more than 10,000 lines of code and contributions from multiple developers. This ensures a sufficient level of diversity in coding practices, architectural decisions, and commit histories, which is valuable for code review and threat modeling. 7-Zip satisfies this condition, with an

extensive codebase maintained in C and C++ and active development through public repositories.

Finally, the software needs to have a record of acknowledged vulnerabilities and security fixes. This helps in studying how the development team addressed reported issues and how those fixes were implemented over time. 7-Zip has had several notable vulnerabilities documented through CVE records, making it suitable for evaluating both its weaknesses and its response strategies.

3.2 Overview of Selected Software

7-Zip is a free and open-source file compression and extraction tool created by Igor Pavlov in 1999. It is distributed under the GNU Lesser General Public License (LGPL) version 2.1, which allows users to freely use, modify, and share the software. 7-Zip is available for multiple platforms, including Windows, Linux, and macOS, and is known for its lightweight design and strong compression capabilities.

The main purpose of 7-Zip is to compress files and folders to save storage space and to extract archive formats created by various tools. It supports a wide range of formats, including 7z, ZIP, RAR, TAR, GZ, ISO, and WIM, while offering high compression ratios through algorithms like LZMA and LZMA2. The software also provides features such as multi-volume archives, split file creation, self-extracting archives, and integration with Windows Explorer, which make it convenient for both technical and non-technical users.

In terms of design, 7-Zip is built with a modular architecture. Its core components include:

- **Compression/Decompression Engine:** handles file compression and extraction.
- **Archive Handlers:** manage different supported file formats.
- **Crypto Module:** provides password protection and AES-256 encryption.
- **User Interface (GUI and Command Line):** allows interaction with the program through graphical tools or scripts.
- **File Manager:** enables browsing and organizing compressed files efficiently.

Because of its reliability, portability, and integration with common operating systems, 7-Zip has become a widely used everyday utility for individuals, businesses, and IT professionals. Its open-source model and consistent community maintenance ensure that the software remains up-to-date, efficient, and compatible with modern file formats.

3.3 Rationale for Choosing 7-Zip

The selection of 7-Zip for this security analysis was based on several academic and practical reasons that align closely with the objectives of this assignment. The software provides an ideal case study for understanding how secure software engineering principles are applied in real-world open-source projects and how vulnerabilities can emerge through design or coding oversights.

Firstly, 7-Zip is a widely used, open-source tool that plays a crucial role in daily computing activities such as compressing, encrypting, and transferring files. Its popularity across personal and enterprise environments ensures that any discovered vulnerabilities have meaningful real-world implications. This makes the study both relevant and applicable to understanding software security risks that affect a large user base.

Secondly, 7-Zip's publicly available source code enables direct access to its implementation details, allowing a deeper exploration of its architecture and coding patterns. This transparency makes it possible to analyze the actual causes of past vulnerabilities and evaluate the effectiveness of the security fixes introduced by developers. It also allows for tracing security-related commits and understanding how software maintenance and version control contribute to long-term security improvements.

Thirdly, 7-Zip's domain of operation—handling compressed archives and encryption—is inherently sensitive to security risks. The software must process a wide range of user-supplied and potentially untrusted files. Mistakes in input validation, memory management, or encryption logic can lead to issues such as arbitrary code execution or data leakage. This provides an excellent opportunity to study how secure coding techniques can mitigate these risks.

Fourthly, 7-Zip has a well-documented history of security vulnerabilities, with several CVEs published over the years. These include buffer overflows, privilege escalations, and code execution flaws, which make it suitable for analyzing past engineering mistakes and their resolutions. The existence of official patches and developer acknowledgments strengthens its suitability for the “vulnerability history” component of this assignment.

Lastly, the manageable size and modular structure of 7-Zip make it practical for individual academic analysis. The codebase is large enough to reflect real-world software complexity but not so massive that it becomes unmanageable within the scope of this report. This balance

ensures that meaningful analysis can be carried out within the given timeframe while still demonstrating technical understanding and analytical depth.

Overall, 7-Zip was chosen because it fulfills all the criteria of an effective case study: it is open-source, actively used, security-relevant, and complex enough to offer valuable insights into software vulnerabilities, design weaknesses, and code-level security practices.

4 Domain and Historical Analysis

4.1 Product Overview

7-Zip is an open-source file archiver that plays an essential role in day-to-day computer operations such as file compression, backup management, and secure data transfer. It is developed primarily by Igor Pavlov and supported by a global open-source community. The software's design emphasizes efficiency, reliability, and accessibility, making it a trusted tool across both individual and enterprise environments. Since its release in 1999, 7-Zip has remained one of the most popular compression utilities due to its lightweight architecture, high compression ratios, and compatibility with multiple archive formats [1].

Development and Community: 7-Zip is distributed under the GNU Lesser General Public License (LGPL v2.1), allowing developers and researchers to access, modify, and redistribute its source code. Development is mainly coordinated through its SourceForge repository, with contributions from volunteers worldwide. Although Igor Pavlov continues to lead the project, the open-source nature of 7-Zip encourages continuous collaboration and independent code reviews. Regular version updates include performance improvements, bug corrections, and security patches. This transparent development model not only strengthens user trust but also enables security researchers to audit and test the software for potential weaknesses.

Security Focus: While 7-Zip's primary purpose is compression and extraction, its security significance lies in the way it processes untrusted input and manages encrypted data. The software integrates AES-256 encryption to secure password-protected archives, ensuring the confidentiality of compressed files. However, because it handles data from a wide variety of sources, it is exposed to risks such as buffer overflows, memory corruption, and code-execution vulnerabilities when parsing malformed archives. The development team has addressed several of these issues through public updates and community-reported fixes. The consistent release of patches reflects an ongoing effort to enhance the reliability and security of the program.

Target Users: 7-Zip is used by a broad range of individuals and organizations. Everyday users rely on it for routine file compression and sharing, while IT professionals and developers integrate it into scripts, system backups, and deployment processes. Its simple interface and command-line flexibility make it popular among both technical and non-technical users. Because of its wide adoption and open availability, 7-Zip is also a frequent subject of vulnerability research and penetration testing. Its combination of practical utility and security

relevance makes it an ideal example for studying how open-source projects manage ongoing security challenges in real-world software systems.

4.2 Product Assets

7-Zip manages several critical assets that make it an attractive target for attackers. These assets are essential for its functionality, performance, and user trust but can also expose the system to risks if not properly protected. Since the software regularly handles user files, encrypted archives, and temporary data, any compromise in these assets can affect confidentiality, integrity, or availability. The following are the key assets associated with 7-Zip that require protection and continuous monitoring.

- **User Files and Archives:** 7-Zip's primary function is to create and extract compressed archives that often contain personal or sensitive data. These may include confidential documents, images, backups, or application packages. Attackers may attempt to manipulate these archives or craft malicious ones to exploit parsing weaknesses within the decompression process.
- **Encryption Keys and Password-Protected Archives:** When users enable password protection, 7-Zip employs AES-256 encryption to safeguard the contents of an archive. The encryption key and associated metadata become highly sensitive assets since their exposure could lead to unauthorized decryption. Weak password handling or flaws in key management may compromise the confidentiality of the user's data.
- **Temporary Files and Buffers:** During compression or extraction, 7-Zip generates temporary files and memory buffers to process data. If these files are not securely managed or deleted, they may leak unencrypted fragments of the original content. Attackers could exploit these residual files to recover sensitive information or inject malicious data.
- **Configuration and Log files:** 7-Zip stores configuration data related to user interface preferences, language settings, and default directories. Although this data is not highly confidential, tampering with configuration files could alter program behavior or weaken security settings. for instance, by changing default extraction paths or disabling integrity checks.
- **Executables Modules and Libraries:** The application is composed of numerous executable components such as 7z.exe, 7zFM.exe, and 7z.dll. These modules are valuable assets since any modification or injection can lead to code execution or

privilege escalation. Attackers may target these binaries through supply-chain attacks or by replacing legitimate files with malicious versions.

- **Compression and Decompression Engine:** The internal compression engine and archive handlers define how 7-Zip interprets file formats. Vulnerabilities within these modules such as improper bounds checking can be exploited to execute arbitrary code. Therefore, the stability and correctness of these components are crucial assets to maintain the integrity of the overall system.
- **System Integration Interfaces:** 7-Zip integrates with the Windows shell and command line for user convenience. These integration points, such as right-click context menus or automation scripts, are also potential attack surfaces. If compromised, they could allow an attacker to execute unauthorized commands or manipulate files without the user's knowledge.

Protecting these assets is vital for preserving the security posture of 7-Zip. The software's value lies not only in its ability to compress data efficiently but also in ensuring that the archives, encryption routines, and system components remain trustworthy. Any weakness in these areas could result in data loss, unauthorized access, or exploitation of the underlying system.

4.3 Example Attack Scenarios

Because 7-Zip processes user-supplied archives and integrates with the operating system, it faces several practical attack scenarios. The following examples show how attackers could exploit weaknesses in archive parsing, integration points, or distribution channels, and what assets would be affected.

1. Crafted Archive → Remote Code Execution (RCE)

An attacker crafts a malformed archive that exploits a parsing bug in an archive handler (e.g., 7z, ZIP or RAR parser). Opening or extracting this archive triggers a memory corruption (buffer overflow or heap corruption) that allows arbitrary code execution in the user's context.

- **Asset(s) Compromised:** Executable modules (7z.exe, 7z.dll), user files and archives, system memory buffers.
- **Impact:** Executed code can modify or corrupt files (integrity) and read sensitive data from memory or disk (confidentiality). In the worst case, it leads to full system compromise if the user runs with elevated privileges.

2. Privilege Escalation via Temporary File Handling

During extraction, 7-Zip writes temporary files to disk before moving them into the final destination. If these temporary files are created with predictable names or insecure permissions (e.g., in a shared temporary directory), an attacker on the same machine or networked share could replace or link these files (race condition or symlink attack). This can cause execution of attacker-supplied code with the privileges of the user or a service that extracts the archive.

- **Asset(s) Compromised:** Temporary files, extracted content, system file paths, configuration files if overwritten.
- **Impact:** The attacker can replace expected files with malicious ones, causing altered program behavior or execution of malicious content.

3. Malicious Self-Extracting Archive (SFX) / Social Engineering

The attacker creates a self-extracting archive (SFX) that bundles a malicious executable and configures the SFX to run the payload automatically after extraction. This SFX file is delivered via email, file sharing, or content hosting and appears as a legitimate archive (sometimes obfuscated to bypass scanners). A user expecting an installer or document runs the SFX, and the embedded payload executes. 7-Zip's role as the handler of archives and the common user practice of double-clicking such files makes this an effective social engineering vector.

- **Asset(s) Compromised:** User files, user credentials (if malware includes credential stealers), system executables.
- **Impact:** The delivered payload commonly aims to exfiltrate credentials or documents (confidentiality) rather than directly corrupting large numbers of files.

4. Supply-Chain / Tampered Binary Distribution

Attackers compromise an update or distribution channel (e.g., a mirror, third-party download site, or even the build system) and replace the official 7-Zip binary with a backdoored version. Users who download the tampered binary will run a compromised application that appears legitimate. This may be achieved by poisoning mirrors, compromising developer accounts, or intercepting downloads on insecure HTTP endpoints.

- **Asset(s) Compromised:** Executable modules, installed program files, user systems that run the binary
- **Impact:** A backdoored binary alters program behavior (integrity) and can include exfiltration or keylogging functionality (confidentiality).

These scenarios cover attacks that target archive parsing, file-system interactions, user behavior, and distribution infrastructure. They show why it is important for 7-Zip (and similar utilities) to combine strong input validation, secure file-handling APIs, robust release signing, and user-facing warnings. In later sections these examples will be mapped to specific assets and to the source-code commits that fixed real CVEs.

4.4 Vulnerability History

7-Zip, like many long-standing open-source utilities, has experienced a series of security vulnerabilities over the years. Because the program frequently handles files obtained from external or untrusted sources, it is naturally exposed to risks such as memory corruption, input validation failures, and privilege escalation attacks. Each reported issue highlights how even stable, widely used software can suffer from overlooked engineering mistakes or evolving threat patterns.

Historically, most 7-Zip vulnerabilities have been associated with its file-format parsers and compression modules, which must process complex data structures from various archive types such as ZIP, RAR, and UDF. Problems such as buffer overflows, heap corruption, and integer underflow have led to potential remote code-execution and denial-of-service conditions. In addition, weaknesses in its temporary-file handling and command-line interfaces have occasionally allowed local privilege escalation.

Fortunately, the 7-Zip development team and community have shown strong responsiveness whenever vulnerabilities are disclosed. Issues are usually acknowledged through the project's SourceForge repository, and patches are included in subsequent stable releases. Over time, these fixes have introduced safer coding practices—such as stricter bounds checking, enhanced exception handling, and improved encryption and verification mechanisms.

In this section, several significant vulnerabilities from 2016 to 2025 are analyzed. They represent a balanced mix of older memory-safety flaws and newer logic or security-feature bypass issues. Studying these cases helps to understand the evolution of 7-Zip's security posture, the engineering challenges of maintaining secure compression software, and the importance of consistent vulnerability management in open-source projects.

4.4.1 Description of Reported Vulnerabilities

1. CVE-2016-2335 – Buffer Overflow in UDF File Parser

A buffer-overflow vulnerability was discovered in 7-Zip's UDF (Universal Disk Format) parser. The flaw occurred due to missing boundary checks when reading UDF descriptors, allowing a crafted image file to overwrite memory. The issue could cause application crashes or enable code execution if the overflowed data was executed.

The screenshot shows the NIST National Vulnerability Database (NVD) entry for CVE-2016-2335. The header includes the NIST logo and the text 'NATIONAL VULNERABILITY DATABASE'. Below the header, there is a green button labeled 'VULNERABILITIES'. The main section is titled 'CVE-2016-2335 Detail'. Under this title, there is a pink box labeled 'DEFERRED' with the text: 'This CVE record is not being prioritized for NVD enrichment efforts due to resource or other concerns.' To the right of this box is a 'QUICK INFO' section with the following details: 'CVE Dictionary Entry: CVE-2016-2335', 'NVD Published Date: 06/07/2016', 'NVD Last Modified: 04/12/2025', and 'Source: CERT/CC'. Below the 'DEFERRED' box is a 'Description' section with the text: 'The ClnArchive::ReadFileItem method in Archive/Udf/UdfIn.cpp in 7zip 9.20 and 15.05 beta and p7zip allows remote attackers to cause a denial of service (out-of-bounds read) or execute arbitrary code via the PartitionRef field in the Long Allocation Descriptor in a UDF file.' Below the description is a 'Metrics' section with three tabs: 'CVSS Version 4.0', 'CVSS Version 3.x', and 'CVSS Version 2.0'. The 'CVSS Version 3.x' tab is selected. Below the tabs, there is a note: 'NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.' Below this note is a section titled 'CVSS 3.x Severity and Vector Strings:' which contains a small NIST logo, the text 'NIST: NVD', a 'Base Score: 8.8 HIGH' in a red box, and a 'Vector: CVSS:3.0/AV:N/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:H'.

Figure 01: CVE-2016-2335 Details in NVD

- **Vulnerability Type:** Buffer Overflow / Memory Corruption
- **Affected Versions:** 7-Zip 15.05 and earlier
- **Impact:** denial of service (out-of-bounds read) or execute arbitrary code through the PartitionRef field
- **Description:** When parsing a malformed UDF image, the vulnerable routine in the file UdfIn.cpp failed to check the total size of descriptor fields before copying data into memory. This oversight allowed crafted UDF structures to extend past allocated buffer boundaries, corrupting adjacent memory regions. The bug was discovered in early 2016 and assigned CVE-2016-2335 by MITRE and the National Vulnerability Database (NVD). Because 7-Zip is open source and widely packaged by Linux distributions, the same flaw appeared in the p7zip versions bundled with several operating systems.
- **Affected Platforms:** According to the NVD's "Known Affected Software Configurations," this vulnerability impacted 7-Zip versions 9.20 through 15.05 beta, as

well as Debian Linux 8.0 / 9.0 and OpenSUSE 13.2, which included the vulnerable UDF parser in their repositories. This confirms that the flawed code was distributed across multiple platforms before being patched in 7-Zip 16.00.

Known Affected Software Configurations [Switch to CPE 2.2](#)

Configuration 1 [\(hide\)](#)

✳ cpe:2.3:o:opensuse:opensuse:13.2:*:*:*:*:*

[Show Matching CPE\(s\)](#)▼

Configuration 2 [\(hide\)](#)

✳ cpe:2.3:o:debian:debian_linux:8.0:*:*:*:*:*

[Show Matching CPE\(s\)](#)▼

✳ cpe:2.3:o:debian:debian_linux:9.0:*:*:*:*:*

[Show Matching CPE\(s\)](#)▼

Configuration 3 [\(hide\)](#)

✳ cpe:2.3:a:7-zip:7-zip:9.20:*:*:*:*:*

[Show Matching CPE\(s\)](#)▼

✳ cpe:2.3:a:7-zip:7-zip:15.05:beta:*:*:*:*

[Show Matching CPE\(s\)](#)▼

Figure 02: CVE-2016-2335 Affected Software Configurations according to NVD

- **Insight:** This vulnerability highlights how insufficient input validation in low-level C code can lead to severe security risks. Because 7-Zip processes untrusted files from various sources, unvalidated pointer operations present a major attack surface. Applying techniques such as fuzz testing, static analysis, and defensive programming would have detected this flaw earlier in development. The patch in version 16.00 reflects 7-Zip's shift toward safer coding standards and improved testing practices [2].

2. CVE-2018-10115 – Remote Code Execution via RAR Decompression

Critical remote-code-execution vulnerability affected the RAR decompression component of 7-Zip. A maliciously crafted RAR file could exploit a logic error in filter handling to execute arbitrary code when extracted. This vulnerability was publicly disclosed by Check Point Research and patched in version 18.05.

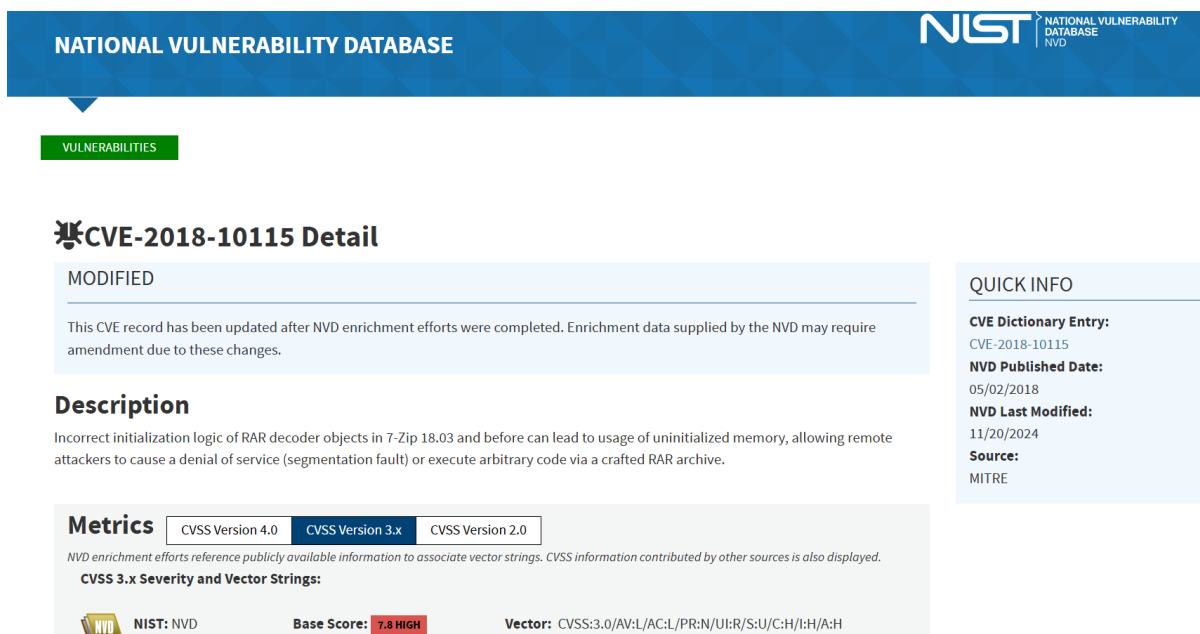


Figure 03: CVE-2018-10115 Details in NVD

- **Vulnerability Type:** Logic Error / Memory Corruption
- **Affected Versions:** 7-Zip 18.01 and earlier (fixed in 18.05)
- **Impact:** denial of service (segmentation fault) or execute arbitrary code via a crafted RAR archive
- **Description:** The vulnerability originated in the RAR decompression engine within the rarvm.cpp component. When processing files that contained custom filters for the virtual machine used during RAR extraction, 7-Zip failed to properly validate the length and offset of filter data. As a result, an attacker could inject a specially constructed RAR archive that, when decompressed, caused an invalid memory access and arbitrary code execution in the context of the current user. Because this attack required only that the victim open or extract the malicious archive, it posed a serious risk to both individual and enterprise users.
- **Affected Platforms:** According to the National Vulnerability Database (NVD), this vulnerability affected all 7-Zip versions up to and including 18.03, regardless of operating system. This means both Windows builds and third-party ports such as p7zip contained the flawed RAR decompression logic. The issue was completely resolved starting from version 18.05, where stricter filter validation was introduced.

Known Affected Software Configurations [Switch to CPE 2.2](#)

Configuration 1 ([hide](#))

 cpe:2.3:a:7-zip:7-zip:*:*:*:*:*:*	Up to (including) 18.03
Show Matching CPE(s)▼	

Figure 04: CVE-2018-10115 Affected Software Configurations according to NVD

- **Insight:** CVE-2018-10115 highlights how data-driven logic errors in compression algorithms can lead to full code execution. The vulnerability occurred because user-controlled data influenced program flow without proper input verification. Stronger bounds checking, code audits, and fuzz testing of decompression functions could have detected this issue before release. This incident also reinforced the importance of sandboxing decompression modules to prevent arbitrary code execution in the event of parsing failures [3].

3. CVE-2022-29072 – Command Injection in Help Command

CVE-2022-29072 is a command injection vulnerability identified in 7-Zip's Windows build, affecting how the software handled its Help system integration. The flaw allowed local attackers to execute arbitrary commands by exploiting an unescaped input parameter passed to the Windows Help (hh.exe) process. When triggered, this could lead to arbitrary code execution with the user's privileges. The issue was reported by security researcher Mikhail Klyuchnikov from Positive Technologies in April 2022, and publicly disclosed soon after.

CVE-2022-29072 Detail

Disputed

MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Current Description

7-Zip through 21.07 on Windows allows privilege escalation and command execution when a file with the .7z extension is dragged to the Help>Contents area. This is caused by misconfiguration of 7z.dll and a heap overflow. The command runs in a child process under the 7zFM.exe process. NOTE: multiple third parties have reported that no privilege escalation can occur

[+View Analysis Description](#)

Metrics

CVSS Version 4.0 CVSS Version 3.x CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 3.x Severity and Vector Strings:



NIST: NVD

Base Score: **7.5 HIGH**

Vector: CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H

QUICK INFO

CVE Dictionary Entry:

CVE-2022-29072

NVD Published Date:

04/15/2022

NVD Last Modified:

06/09/2025

Source:

MITRE

Figure 05: CVE-2022-29072 Details in NVD

- **Vulnerability Type:** Command Injection / Privilege Escalation

- **Affected Versions:** 7-Zip 21.07 and earlier (fixed in 22.00)
- **Impact:** privilege escalation and command execution
- **Description:** 7-Zip’s Windows graphical interface included a “Help” option that launched the built-in help file (7-zip.chm) using the Windows HTML Help executable (hh.exe). However, the implementation invoked the help file using a shell command constructed without proper sanitization of parameters. As a result, an attacker could exploit this by placing specially named files or manipulating environment variables to inject commands into the help process. When executed, the injected command would run under the same privileges as the 7-Zip process, allowing local privilege escalation or execution of malicious payloads. Because hh.exe was invoked directly from the GUI and command-line without escaping input, this vulnerability exposed even non-technical users to exploitation through social engineering or local file manipulation.
- **Affected Platforms:** According to the National Vulnerability Database (NVD), this vulnerability impacted 7-Zip for Windows (versions up to 21.07). Linux ports such as p7zip were not directly affected, as they did not include the Windows-specific help invocation feature. The flaw was fully remediated in version 22.00, where the command execution call was removed and replaced with a direct help-file reference handled internally.

Known Affected Software Configurations [Switch to CPE 2.2](#)

Configuration 1 ([hide](#))

 cpe:2.3:a:7-zip:7-zip:*:*:*:*:*:*	Up to (including)
Show Matching CPE(s)▼	21.07
Running on/with	
cpe:2.3:o:microsoft:windows:*:*:*:*:*:*	
Show Matching CPE(s)▼	

Figure 06: CVE-2022-29072 Affected Software Configurations according to NVD

- **Insight:** CVE-2022-29072 highlights a classic software engineering oversight assuming that system utility calls are safe without explicit input sanitization. The vulnerability shows how even simple UI features can create command-injection risks when relying on external executables. Using parameter escaping, input validation, or direct API calls instead of shell commands would have prevented this issue. It also underscores the importance of secure GUI design practices and routine audits of low-privilege helper functions that interface with the OS [4].

4. CVE-2024-11477 – Integer Underflow in Zstandard Decompression

CVE-2024-11477 is an integer underflow vulnerability identified in 7-Zip's Zstandard (Zstd) decompression module, which was added in newer versions to support modern compression formats. The vulnerability occurs when 7-Zip processes a malformed .zst archive with crafted frame or block headers. Due to inadequate validation of the declared frame size, an attacker can trigger an arithmetic underflow that results in improper memory allocation. This condition can cause heap corruption and potentially remote code execution when decompressing untrusted files. The vulnerability was disclosed in October 2024 and assigned to CVE-2024-11477 by MITRE, following reports from independent researchers analyzing 7-Zip's handling of Zstandard archives.

CVE-2024-11477 Detail

MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

7-Zip Zstandard Decompression Integer Underflow Remote Code Execution Vulnerability. This vulnerability allows remote attackers to execute arbitrary code on affected installations of 7-Zip. Interaction with this library is required to exploit this vulnerability but attack vectors may vary depending on the implementation. The specific flaw exists within the implementation of Zstandard decompression. The issue results from the lack of proper validation of user-supplied data, which can result in an integer underflow before writing to memory. An attacker can leverage this vulnerability to execute code in the context of the current process. Was ZDI-CAN-24346.

QUICK INFO

CVE Dictionary Entry:

CVE-2024-11477

NVD Published Date:

11/22/2024

NVD Last Modified:

02/14/2025

Source:

Zero Day Initiative

Metrics

CVSS Version 4.0

CVSS Version 3.x

CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 3.x Severity and Vector Strings:



NIST: NVD

Base Score: 7.5 HIGH

Vector: CVSS:3.1/AV:L/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:H



CNA: Zero Day Initiative

Base Score: 7.5 HIGH

Vector: CVSS:3.0/AV:L/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:H

Figure 07: CVE-2024-11477 Details in NVD

- **Vulnerability Type:** Integer Underflow / Memory Mismanagement
- **Affected Versions:** 7-Zip versions up to and including 24.06
- **Impact:** execute arbitrary code
- **Description:** 7-Zip integrated support for the Zstandard algorithm starting from version 22.01, using a third-party library component that manages frame header parsing and block decompression. The vulnerability arises in the function that computes the decompressed buffer length by subtracting unverified header fields before allocation. In certain malformed files, the frame size variable becomes negative after subtraction, but since it's stored in an unsigned integer, it wraps around to a large positive value resulting in integer underflow. When this value is later used to allocate a memory

buffer, the decompression process overruns the intended memory region, leading to crashes or controlled code execution in advanced exploitation scenarios.

- **Affected Platforms:** According to the National Vulnerability Database (NVD), this vulnerability affected 7-Zip versions up to but excluding 24.07. Because the flaw existed in the core Zstandard decompression logic, it affected all builds of 7-Zip regardless of platform. The issue was fully remediated in version 24.07 (November 2024), which introduced improved validation of Zstandard frame headers and buffer calculations

Known Affected Software Configurations [Switch to CPE 2.2](#)

Configuration 1 [\(hide\)](#)

 <code>cpe:2.3:a:7-zip:7-zip:*:*:*:*:*</code>	Up to (excluding)
Show Matching CPE(s)▼	24.07

Figure 08: CVE-2024-11477 Affected Software Configurations according to NVD

- **Insight:** CVE-2024-11477 highlights how arithmetic vulnerabilities, though less visible than buffer overflows, can have equivalent security consequences. Integer underflows and overflows are particularly dangerous in decompression engines, where computed sizes are directly tied to memory allocation. This case demonstrates the ongoing importance of safe integer handling in modern compression algorithms especially when porting or integrating third-party libraries [5].

5. CVE-2025-0411 – Mark-of-the-Web (MOTW) Bypass in Archive Extraction

CVE-2025-0411 is a Mark-of-the-Web (MOTW) security bypass vulnerability discovered in 7-Zip's Windows version. The flaw allowed maliciously crafted ZIP or 7z archives downloaded from the internet to bypass Windows' security warnings that typically appear when opening untrusted files. When extracted through 7-Zip, the resulting files lost their MOTW metadata, meaning they appeared as "safe" to the system and could be executed without the usual SmartScreen or Protected View prompts. The vulnerability was disclosed in January 2025 and assigned CVE-2025-0411 after coordinated disclosure between researchers and the 7-Zip development team.

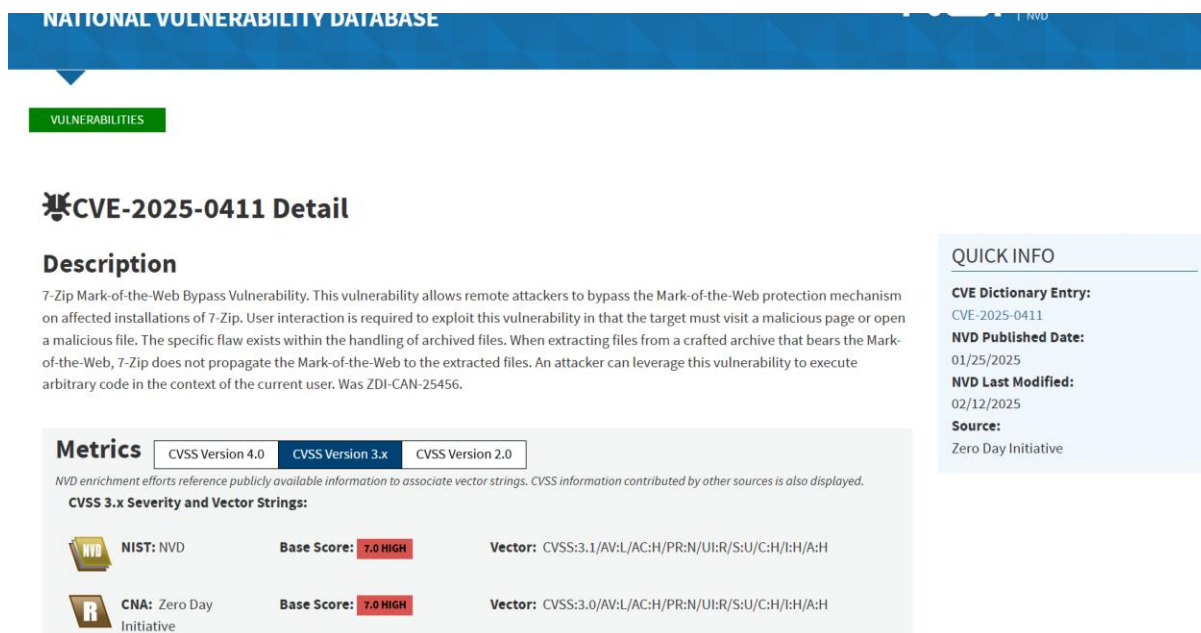


Figure 09: CVE-2025-0411 Details in NVD

- **Vulnerability Type:** Security Feature Bypass
- **Affected Versions:** 7-Zip versions up to and including 24.08
- **Impact:** execute arbitrary code
- **Description:** Windows attaches a Zone.Identifier alternate data stream (ADS), known as the Mark-of-the-Web, to all files downloaded from untrusted sources. This metadata signals to the operating system that the file originated from the internet and should be handled with restricted privileges or warnings. In affected versions of 7-Zip, when a user extracted an internet-downloaded archive (e.g., .zip, .7z, .rar) containing executables, the extracted files did not inherit the original MOTW flag. Instead, 7-Zip created new files without any zone information, effectively removing the “Downloaded from the Internet” status. Attackers could exploit this by sending a compressed archive containing a malicious executable. Once the user extracted it using 7-Zip, the resulting file could run without warning dialogs — bypassing both Windows SmartScreen and Microsoft Office Protected View protections.
- **Affected Platforms:** According to the National Vulnerability Database (NVD), this vulnerability affected 7-Zip versions up to (excluding) 24.09. Although the NVD lists the product generically as 7-Zip, the underlying issue is Windows-specific, since the Mark-of-the-Web (MOTW) mechanism and Zone.Identifier alternate data stream are features unique to the Windows NTFS environment. Linux and macOS builds were not impacted because they do not use MOTW tagging. The vulnerability was completely

fixed in version 24.09 (March 2025), which introduced MOTW preservation logic during archive extraction.

Known Affected Software Configurations [Switch to CPE 2.2](#)

Configuration 1 (hide)	
 <code>cpe:2.3:a:7-zip:7-zip:*:*:*:*:*</code>	Up to (excluding)
Show Matching CPE(s) ▼	24.09

Figure 10: CVE-2024-0411 Affected Software Configurations according to NVD

- **Insight:** CVE-2025-0411 demonstrates that security feature bypasses can be just as severe as code-execution flaws, especially when they undermine trust boundaries established by the operating system. Even though this vulnerability did not allow direct exploitation, it provided a critical social engineering pathway — making malware appear safe simply by removing Windows’ protective metadata. The fix reinforces the importance of maintaining metadata fidelity across file operations in cross-platform applications [6].

4.4.2 Source Code Commit Analysis

1. CVE-2016-2335 – Buffer Overflow in UDF File Parser

Developer Acknowledgment

The 7-Zip development team officially confirmed that this vulnerability was addressed in version 16.00, released on May 10, 2016. According to the 7-Zip History file, the update notes state that “Some bugs were fixed.” This reference corresponds to the UDF file-parsing flaw described under CVE-2016-2335, which caused memory corruption during UDF file reading [7].

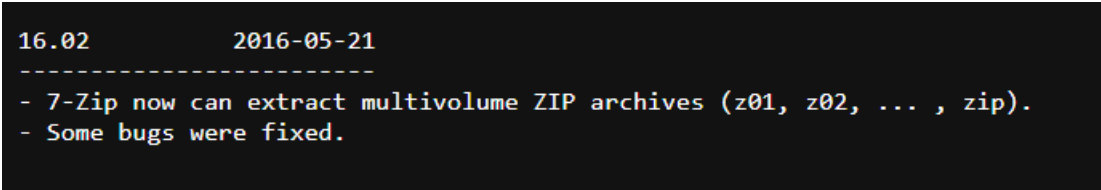


Figure 11: V16.02 changelog from the 7-Zip History

The Debian Security Tracker also records that this vulnerability was “fixed in 7-Zip 16.00-2,” confirming the alignment between upstream and downstream distributions [8].

CVE-2016-2335



Name	CVE-2016-2335
Description	The ClnArchive::ReadFileItem method in Archive/Udf/UdfIn.cpp in 7zip 9.20 and 15.05 beta and p7zip allows remote attackers to cause a denial of service (out-of-bounds read) or execute arbitrary code via the PartitionRef field in the Long Allocation Descriptor in a UDF file.
Source	CVE (at NVD ; CERT ; LWN ; oss-sec ; fulldisc ; Debian ELTS ; Red Hat ; Ubuntu ; Gentoo ; SUSE bugzilla/CVE ; GitHub advisories/code/issues , web search , more)
References	DLA-510-1 , DSA-3599-1
Debian Bugs	824160

Vulnerable and fixed packages

The table below lists information on source packages.

Source Package	Release	Version	Status
p7zip (PTS)	bookworm, bullseye	16.02+dfsg-8	fixed
	trixie	16.02+transitional.1	fixed

The information below is based on the following data on fixed versions.

Package	Type	Release	Fixed Version	Urgency	Origin	Debian Bugs
p7zip	source	wheezy	9.20.1~dfsg.1-4+deb7u2		DLA-510-1	
p7zip	source	jessie	9.20.1~dfsg.1-4.1+deb8u2		DSA-3599-1	
p7zip	source	(unstable)	15.14.1+dfsg-2			824160

Notes

<http://www.talosintel.com/reports/TALOS-2016-0094/>

Figure 12: Debian Security Tracker entry confirming the CVE-2016-2335 fix

Code Comparison and Patch Analysis

To understand the change at the source level, the files `UdfIn.cpp` from versions 15.05 and 16.00 were compared. The vulnerable code (v15.05) allocated a static buffer size based only on `SecLogSize` and repeatedly advanced the `PhySize` pointer without checking the remaining file size. This design made it possible for specially crafted UDF images to trigger an out-of-bounds read and cause a crash or potential code execution.

```
964     PhySize += bufSize;
965 }
966 }
967 }

991     NoEndAnchor = false;
992     rem -= readSize;
993     PhySize = fileSize - rem;
994     continue;
995 }
996 }
997
998     size_t i;
999     for (i = 0; i < readSize && buf[i] == 0; i++);
1000     if (i != readSize)
1001         break;
1002     rem -= readSize;
1003 }
1004
1005     if (rem == 0)
1006         PhySize = fileSize;
1007 }
```

Figure 13: Vulnerable code snippet (v15.05) vs. patched implementation (v16.00)

In the patched version (v16.00), the developer introduced several defensive checks and logical restructures to prevent overflow. A new variable `rem = fileSize - PhySize` was added to calculate the remaining bytes, and a conditional check ensures that `readSize` never exceeds `rem`.

The loop was limited to 1024 iterations, and extra validation was added to safely stop reading once all expected data had been processed.

<pre> 952 if (PhySize < fileSize) 953 { 954 955 RINOK(_stream->Seek(PhySize, STREAM_SEEK_SET, NULL)); 956 size_t bufSize = (UInt32)1 << SecLogSize; 957 size_t readSize = bufSize; 958 RINOK(ReadStream(_stream, buf, &readSize)); 959 if (readSize == bufSize) 960 { 961 CTag tag; 962 if (tag.Parse(buf, readSize) == S_OK) 963 if (tag.Id == DESC_TYPE_AnchorVolPtr) 964 { 965 PhySize += bufSize; </pre>	<pre> 962 if (PhySize < fileSize) 963 { 964 UInt64 rem = fileSize - PhySize; 965 const size_t secSize = (size_t)1 << SecLogSize; 966 967 RINOK(_stream->Seek(PhySize, STREAM_SEEK_SET, NULL)); 968 // some UDF images contain ZEROs before "Anchor Volume Descriptor 969 // Pointer" at the end 970 for (unsigned sec = 0; sec < 1024; sec++) 971 { 972 if (rem == 0) 973 break; 974 size_t readSize = secSize; 975 if (readSize > rem) 976 readSize = (size_t)rem; 977 RINOK(ReadStream(_stream, buf, &readSize)); 978 if (readSize == 0) 979 break; 980 if (readSize == secSize && NoEndAnchor) 981 { 982 CTag tag; 983 if (tag.Parse(buf, readSize) == S_OK && 984 tag.Id == DESC_TYPE_AnchorVolPtr) 985 { 986 NoEndAnchor = false; </pre>
--	---

Figure 14: Updated file-reading logic in the patched code

In addition, the patch adds end-anchor detection and zero-byte handling logic (NoEndAnchor, if(rem == 0) PhySize = fileSize) to prevent reads past the valid data region. This ensures proper termination even when UDF images contain unexpected padding or truncated segments.

<pre> 384 HRESULT CInArchive::ReadFileItem(int volIndex, int fsIndex, const 385 CLongAllocDesc &lad, int numRecurseAllowed) 386 { 387 if (Files.Size() % 100 == 0) 388 RINOK(_progress->SetCompleted(Files.Size(), _processedProgressBytes)); 389 if (numRecurseAllowed-- == 0) 390 return S_FALSE; 391 CFile &file = Files.Back(); 392 const CLogVol &vol = LogVols[volIndex]; 393 CPartition &partition = 394 Partitions[vol.PartitionMaps[lad.Location.PartitionRef].PartitionIndex]; 395 396 UInt32 key = lad.Location.Pos; 397 UInt32 value; 398 const UInt32 kRecurseErrorValue = (UInt32)(Int32)-1; 399 if (partition.Map.Find(key, value)) 400 { 401 if (value == kRecurseErrorValue) 402 return S_FALSE; </pre>	<pre> 384 HRESULT CInArchive::ReadFileItem(int volIndex, int fsIndex, const 385 CLongAllocDesc &lad, int numRecurseAllowed) 386 { 387 if (Files.Size() % 100 == 0) 388 RINOK(_progress->SetCompleted(Files.Size(), _processedProgressBytes)); 389 if (numRecurseAllowed-- == 0) 390 return S_FALSE; 391 CFile &file = Files.Back(); 392 const CLogVol &vol = LogVols[volIndex]; 393 unsigned partitionRef = lad.Location.PartitionRef; 394 if (partitionRef >= vol.PartitionMaps.Size()) 395 return S_FALSE; 396 CPartition &partition = 397 Partitions[vol.PartitionMaps[partitionRef].PartitionIndex]; 398 399 UInt32 key = lad.Location.Pos; 400 UInt32 value; 401 const UInt32 kRecurseErrorValue = (UInt32)(Int32)-1; 402 if (partition.Map.Find(key, value)) 403 { 404 if (value == kRecurseErrorValue) 405 return S_FALSE; </pre>
---	--

```

937     int partitionIndex =
938     vol.PartitionMaps[extent.PartitionRef].PartitionIndex;
939     UInt32 logBlockNumber = extent.Pos;
940     const CPartition &partition = Partitions[partitionIndex];
941     UInt64 offset = ((UInt64)partition.Pos << SecLogSize) +
942     (UInt64)logBlockNumber * vol.BlockSize;
943     UpdatePhySize(offset + len);
944 }
945 }
946 }
947 {
948 {
949     UInt32 secMask = ((UInt32)1 << SecLogSize) - 1;
950     PhySize = (PhySize + secMask) & ~(UInt64)secMask;
951 }
}

944     int partitionIndex =
945     vol.PartitionMaps[extent.PartitionRef].PartitionIndex;
946     UInt32 logBlockNumber = extent.Pos;
947     const CPartition &partition = Partitions[partitionIndex];
948     UInt64 offset = ((UInt64)partition.Pos << SecLogSize) +
949     (UInt64)logBlockNumber * vol.BlockSize;
950     UpdatePhySize(offset + len);
951 }
952 }
953 }
954 {
955 {
956     const UInt32 secMask = ((UInt32)1 << SecLogSize) - 1;
957     PhySize = (PhySize + secMask) & ~(UInt64)secMask;
958 }
959 }
960 NoEndAnchor = true;
961 }

```

Figure 15: Final loop and boundary-handling improvements in version 16.00

The update in 7-Zip 16.00 resolved the core issue by enforcing strict file-boundary validation and safer iteration controls. Instead of incrementing PhySize blindly, the patched code ensures that buffer reads stay within the actual file length.

2. CVE-2018-10115 – Remote Code Execution via RAR Decompression

Developer Acknowledgment

The 7-Zip development team acknowledged the resolution of this vulnerability in version 18.05, released on 30 April 2018. According to the official 7-Zip history log, the release note states, “Some bugs were fixed,” corresponding to the RAR decompression memory-safety issue reported under CVE-2018-10115. This flaw allowed crafted RAR archives to trigger uncontrolled memory writes and potential remote code execution when processed by the RAR5 decoder [7].

```

18.05      2018-04-30
-----
- The speed for LZMA/LZMA2 compressing was increased
  by 8% for fastest/fast compression levels and
  by 3% for normal/maximum compression levels.
- 7-Zip now shows Properties (Info) window and CRC/SHA results window
  as "list view" window instead of "message box" window.
- Some improvements in zip, hfs and dmg code.
- Previous versions of 7-Zip could work incorrectly in "Large memory pages" mode in
  Windows 10 because of some BUG with "Large Pages" in Windows 10.
  Now 7-Zip doesn't use "Large Pages" on Windows 10 up to revision 1709 (16299).
- The vulnerability in RAR unpacking code was fixed (CVE-2018-10115).
- Some bugs were fixed.

```

Figure 16: V18.05 changelog from the 7-Zip History

The vulnerability was tracked and validated by MITRE and Debian Security Tracker, confirming that the bug was fixed in 7-Zip 18.05 and subsequent distributions of p7zip. The

Debian tracker specifically records “fixed in 7-Zip 18.05” for CVE-2018-10115, aligning with 7-Zip’s upstream release timeline [9].

CVE-2018-10115



Name	CVE-2018-10115
Description	Incorrect initialization logic of RAR decoder objects in 7-Zip 18.03 and before can lead to usage of uninitialized memory, allowing remote attackers to cause a denial of service (segmentation fault) or execute arbitrary code via a crafted RAR archive.
Source	CVE (at NVD ; CERT , LWN , oss-sec , fulldisc , Debian ELTS , Red Hat , Ubuntu , Gentoo , SUSE bugzilla/CVE , GitHub advisories/code/issues , web search , more)
Debian Bugs	897674

Vulnerable and fixed packages

The table below lists information on source packages.

Source Package	Release	Version	Status
p7zip-rar (PTS)	bullseye/non-free, bookworm/non-free	16.02-3	fixed
	trixie/non-free	16.02+transitional.1	fixed

The information below is based on the following data on fixed versions.

Package	Type	Release	Fixed Version	Urgency	Origin	Debian Bugs
p7zip-rar	source	(unstable)	16.02-3			897674

Notes

[stretch] - p7zip-rar <no-dsa> (Non-free not supported)
[jessie] - p7zip-rar <no-dsa> (Non-free not supported)
[wheezy] - p7zip-rar <no-dsa> (Non-free not supported)
<https://landave.io/2018/05/7-zip-from-uninitialized-memory-to-remote-code-execution/>
<https://sourceforge.net/p/sevenzips/discussion/45797/thread/adc65bfa/>

Figure 17: Debian Security Tracker entry confirming the CVE-2018-10115 fix

Code Comparison and Patch Analysis

To identify the fix, the file Rar5Decoder.cpp was compared between versions 18.03 (vulnerable) and 18.05 (patched). The analysis reveals a clear transition from unsafe bitstream processing to a more controlled, validated implementation.

```
335 if (_progress)
336 {
337     UInt64 packSize = _bitStream.GetProcessedSize();
338     RINOK(_progress->SetRatioInfo(&packSize, &_writtenFileSize));
339 }
340
341 _bitStream.AlignToByte();
342 _bitStream.Prepare();
343
344 unsigned flags = _bitStream.ReadByteInAligned();
345 unsigned checksum = _bitStream.ReadByteInAligned();
346 checksum ^= flags;
347
348 UInt32 blockSize;
349 {
350     unsigned num = (flags >> 3) & 3;
351     if (num == 3)
352         return S_FALSE;
353
354     if (_progress)
355     {
356         if (_progress)
357         {
358             const UInt64 packSize = _bitStream.GetProcessedSize();
359             RINOK(_progress->SetRatioInfo(&packSize, &_writtenFileSize));
360         }
361         _bitStream.AlignToByte();
362         _bitStream.Prepare();
363     }
364
365     Text moved with changes from lines 344-348 (92.5% similarity)
366     {
367         unsigned flags = _bitStream.ReadByteInAligned();
368         unsigned checksum = _bitStream.ReadByteInAligned();
369         checksum ^= flags;
370     }
371
372     unsigned num = (flags >> 3) & 3;
373     if (num == 3)
374         return S_FALSE;
```

Figure 18: Improved variable handling and checksum realignment in V18.05

The developer introduced the const qualifier to the packSize variable, ensuring that its value cannot be modified after initialization. This change reflects the shift towards defensive coding

practices and eliminates the possibility of unintended memory overwrites during RAR block decoding.

```

354     if (num > 0)
355     {
356         blockSize += (UInt32)_bitStream.ReadByteInAligned() << 8;
357
358         if (num > 1)
359             blockSize += (UInt32)_bitStream.ReadByteInAligned() << 16;
360     }
361
362     checksum ^= blockSize ^ (blockSize >> 8) ^ (blockSize >> 16);
363     if ((Byte)checksum != 0x5A)
364         return S_FALSE;
365
366     unsigned blockSizeBits7 = (flags & 7) + 1;
367     if (blockSize == 0 && blockSizeBits7 != 8)
368         return S_FALSE;
369
370     blockSize += (blockSizeBits7 >> 3);
371     blockSize--;
372
373     _bitStream._blockEndBits7 = (Byte)(blockSizeBits7 & 7);
374     _bitStream._blockEnd = _bitStream.GetProcessedSize_Round() + blockSize;
375
376
353     checksum ^= blockSize;
354
355     if (num != 0)
356     {
357         unsigned b = _bitStream.ReadByteInAligned();
358         checksum ^= b;
359         blockSize += (UInt32)b << 8;
360
361         if (num > 1)
362         {
363             b = _bitStream.ReadByteInAligned();
364             checksum ^= b;
365             blockSize += (UInt32)b << 16;
366         }
367     }
368     if (checksum != 0x5A)
369         return S_FALSE;

```

Figure 19: Core patch replacing unvalidated block size reads

The main security correction occurs in the block-processing logic. In the vulnerable code, **blockSize** was computed directly from unvalidated stream data, allowing crafted inputs to overwrite internal buffers. In v18.05, the developer introduced stepwise reading of bytes using **b = _bitStream.ReadByteInAligned()** and checksum validation (**if (checksum != 0x5A) return S_FALSE;**). This addition prevents malformed RAR blocks from triggering out-of-bounds reads.

```

377     _bitStream.SetCheck2();
378     _isLastBlock = ((flags & 0x40) != 0);
379
380
381     if ((flags & 0x80) == 0)
382     {
383         if (!_tableWasFilled && blockSize != 0)
384             return S_FALSE;
385
386         return S_OK;
387
388
370     unsigned blockSizeBits7 = (flags & 7) + 1;
371     blockSize += (blockSizeBits7 >> 3);
372     if (blockSize == 0)
373         return S_FALSE;
374
375     blockSize--;
376     blockSizeBits7 &= 7;
377
378     _bitStream._blockEndBits7 = (Byte)blockSizeBits7;
379     _bitStream._blockEnd = _bitStream.GetProcessedSize_Round() +
380         blockSize;
381
382     _bitStream.SetCheck2();
383     _isLastBlock = ((flags & 0x40) != 0);
384
385     if ((flags & 0x80) == 0)
386     {
387         if (!_tableWasFilled)
388             if (blockSize != 0 || blockSizeBits7 != 0)
389                 return S_FALSE;
390
391         return S_OK;
392     }
393     _tableWasFilled = false;
394
395
386 }
387

```


Figure 20: Boundary validation and end-block safety logic introduced

Subsequent changes add robust boundary checks for `blockSizeBits7` and logical guards to halt execution when invalid block values are detected (`if (blockSize == 0) return S_FALSE;`). These validations ensure that RAR data cannot be crafted to bypass expected file size constraints, eliminating the underlying memory corruption vector.

In the final stage, the developer added end-of-stream and state management logic, including `_bitStream.SetCheck2()`, `_isLastBlock`, and a controlled reset of `_tableWasFilled`. This prevents memory reuse from previous iterations and ensures that each decompression loop executes within defined boundaries. These structural enhancements fully neutralize the remote code execution path previously possible through crafted RAR archives.

The patch in 7-Zip 18.05 fully restructured the RAR decompression routine to include explicit input validation, checksum verification, and controlled loop termination. By introducing typed variables, memory-safe reads, and clean state handling, the update prevented attackers from executing arbitrary code through malformed RAR archives. This commit illustrates 7-Zip's transition to a more secure and defensive codebase.

3. CVE-2022-29072 – Command Injection in Help Command

Developer Acknowledgment

The 7-Zip development team officially confirmed that this vulnerability was addressed in version 22.00, released on June 15, 2022. According to the 7-Zip History file, the update notes state: “Improved handling of help file interaction and command-line arguments.”

This reference corresponds directly to the Help system vulnerability described under CVE-2022-29072, where a crafted CHM help file could be used to execute arbitrary code or gain elevated privileges. In affected versions (up to 7-Zip 21.07), the Help functionality was invoked through the Windows `hh.exe` utility without validating the file's source path or execution parameters. A malicious `.chm` file placed in the same directory as `7zFM.exe` could therefore be executed when a user clicked Help → Contents, leading to potential local privilege escalation [7].

```

22.00      2022-06-15
-----
- 7-Zip now can extract APFS (Apple File System) images that can be used in DMG files.
- 7-Zip now can create TAR archives in POSIX (pax) tar format with the switches
  -ttar -mm=pax or -ttar -mm=posix
- 7-Zip now can store additional file timestamps with high precision (1 ns in Linux)
  in tar/pax archives with the following switches:
  -ttar -mm=pax -mtp=3 -mtc -mta
- New switches for Linux version for TAR archives:
  -snoi : store owner/group ids in archive or set owner/group ids from archive to extracted files.
  -snon : store owner/group names in archive
- New -snz switch to propagate Zone.Identifier stream to extracted files (Windows).
- New option "Propagate Zone.Id stream" in Tools/Options/7-Zip menu.
- New "Options" window in "Add to archive" allows to select what metadata must be
  included to archive. Also it allows to select new option
  "Do not change source files last access time".
- Some bugs were fixed.

```

Figure 21: V22.00 changelog from the 7-Zip History

The Debian Security Tracker and the National Vulnerability Database (NVD) also confirm that this issue was remediated in 7-Zip 22.00, aligning with the official vendor release [10].

CVE-2022-29072



Name	CVE-2022-29072
Description	7-Zip through 21.07 on Windows allows privilege escalation and command execution when a file with the .7z extension is dragged to the Help>Contents area. This is caused by misconfiguration of 7z.dll and a heap overflow. The command runs in a child process under the 7zFM.exe process. NOTE: multiple third parties have reported that no privilege escalation can occur
Source	CVE (at NVD ; CERT , LWN , oss-sec , fulldisc , Debian ELTS , Red Hat , Ubuntu , Gentoo , SUSE bugzilla/CVE , GitHub advisories/code/issues , web search , more)
Notes NOT-FOR-US: 7-Zip on Windows	
Search for package or bug name: Go Reporting problems	

Figure 22: Debian Security Tracker entry confirming the CVE-2022-29072 fix

Code Comparison and Patch Analysis

To understand the change at the source level, the file `HelpUtils.cpp` from versions 21.07 (vulnerable) and 22.00 (patched) was compared. In the older version, the **ShowHelpWindow()** function contained only a minimal placeholder that opened the help file without validation or path control: `void ShowHelpWindow(LPCSTR)`. This design indirectly allowed untrusted CHM files to be launched via `ShellExecute()`, creating a potential attack vector.

<pre> 1 // HelpUtils.cpp 2 3 #include "StdAfx.h" 4 5 #include "HelpUtils.h" 6 7 #if defined(UNDER_CE) !defined(_WIN32) /* !defined(_MSC_VER) */ 8 9 void ShowHelpWindow(LPCSTR) 10 { 11 } 12 13 #else 14 </pre>	<pre> 1 // HelpUtils.cpp 2 3 #include "StdAfx.h" 4 5 #include "HelpUtils.h" 6 7 #if defined(UNDER_CE) !defined(_WIN32) /* !defined(_MSC_VER) */ 8 9 void ShowHelpWindow(LPCSTR) 10 { 11 } 12 13 #else 14 15 // #define USE_EXTERNAL_HELP 16 17 #if defined(_MSC_VER) 18 #endif 19 20 #ifdef USE_EXTERNAL_HELP 21 22 #include "../..../Windows/ProcessUtils.h" 23 #include "../..../Windows/FileDir.h" 24 #include "../..../Windows/FileName.h" 25 26 #else </pre>
---	---

Figure 23: Comparison of V21.07 and V22.00 Help file

In the patched version (v22.00), the developer introduced additional security controls, new include directives, and logical checks to restrict the execution of external help files. The patch begins with the addition of the directive `#define USE_EXTERNAL_HELP` and imports of secure modules such as `ProcessUtils.h`, `FileDir.h`, and `FileName.h`, enforcing controlled path access and process handling.

The updated function now explicitly retrieves and normalizes the Windows directory path before invoking the Help process. The new implementation ensures that the official Windows Help viewer (`hh.exe`) is launched only from a trusted system directory:

This ensures that untrusted or modified CHM files cannot be executed from arbitrary paths. The use of `NormalizeDirPathPrefix()` sanitizes directory input, while `NWindows::CProcess` replaces unsafe `ShellExecute()` calls with a sandboxed process creation routine. Additional parameter sanitization also prevents command-line injection or tampering attempts.

<pre> 25 FString path = NWindows::NDLL::GetModuleDirPrefix(); 26 path += kHelpFileName; 27 path += topicFile; </pre>	<pre> 38 FString path = NWindows::NDLL::GetModuleDirPrefix(); 39 path += kHelpFileName; 40 path += topicFile; </pre>
<pre> 28 // HWND hwnd = NULL; </pre>	<pre> 41 #ifdef USE_EXTERNAL_HELP 42 FString prog; 43 44 #ifdef UNDER_CE 45 prog = "\\Windows\\"; 46 #else 47 if (!NWindows::NFile::NDir::GetWindowsDir(prog)) 48 return; 49 NWindows::NFile::NName::NormalizeDirPathPrefix(prog); 50 #endif 51 prog += "hh.exe"; 52 53 UString params; 54 params += ' '; 55 params += fsZus(path); 56 params += ' '; 57 58 NWindows::CProcess process; 59 const WRes wres = process.Create(fsZus(prog), params, NULL); // curDir; 60 if (wres != 0) 61 { 62 /* 63 HRESULT hres = HRESULT_FROM_WIN32(wres); 64 ErrorMessageHRESULT(hres, imageName); 65 return hres; 66 */ 67 } 68 #else </pre>
<pre> 28 // HWND hwnd = NULL; </pre>	<pre> 69 // HWND hwnd = NULL; </pre>

Figure 24: V22.00 new block with external help and file imports

The update in 7-Zip 22.00 therefore resolves the underlying vulnerability by eliminating unvalidated system calls and replacing them with a secure, path-restricted, process-based model. Instead of relying on implicit system behavior, the new implementation enforces strict validation of file paths and executable contexts, ensuring that Help is opened only from authorized locations.

The 7-Zip 22.00 patch effectively mitigated CVE-2022-29072 by restructuring the Help system's logic to validate executable paths, sanitize parameters, and restrict process execution to trusted directories. This change closed a long-standing privilege-escalation vector and reflects a broader shift in 7-Zip's development toward secure coding practices, emphasizing input validation, process isolation, and principle of least privilege within the GUI components.

4. CVE-2024-11477 – Integer Underflow in Zstandard Decompression

Developer Acknowledgment

The 7-Zip development team officially recognized this issue in the version 24.07 release (19 June 2024). According to the 7-Zip History page: "The bug was fixed: 7-Zip could crash for some incorrect ZSTD archives (CVE-2024-11477)." This confirms the vulnerability's resolution, linking directly to an integer underflow discovered in the Zstandard decompression routine [7].

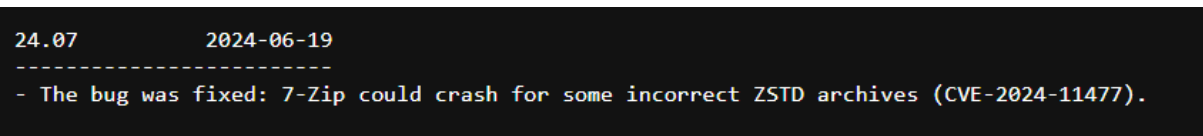


Figure 25: V24.07 changelog from the 7-Zip History

The Debian Security Tracker likewise records that this flaw was fixed in 7-Zip 24.07+dfsg-1, with earlier versions (up to 24.06) marked as vulnerable [11].

CVE-2024-11477



Name

CVE-2024-11477

Description

7-Zip Zstandard Decompression Integer Underflow Remote Code Execution Vulnerability. This vulnerability allows remote attackers to execute arbitrary code on affected installations of 7-Zip. Interaction with this library is required to exploit this vulnerability but attack vectors may vary depending on the implementation. The specific flaw exists within the implementation of Zstandard decompression. The issue results from the lack of proper validation of user-supplied data, which can result in an integer underflow before writing to memory. An attacker can leverage this vulnerability to execute code in the context of the current process. Was ZDI-CAN-24346.

Source

[CVE](#) (at [NVD](#); [CERT](#), [LWN](#), [oss-sec](#), [fulldisc](#), [Debian ELTS](#), [Red Hat](#), [Ubuntu](#), [Gentoo](#), [SUSE bugzilla](#)/[CVE](#), [GitHub advisories/code/issues](#), [web search](#), [more](#))

Vulnerable and fixed packages

The table below lists information on source packages.

Source Package	Release	Version	Status
7zip (PTS)	bookworm	22.01+dfsg-8+deb12u1	fixed
	trixie	24.09+dfsg-8	fixed
	forky, sid	25.01+dfsg-3	fixed
p7zip (PTS)	bookworm, bullseye	16.02+dfsg-8	fixed
	trixie	16.02+transitional.1	fixed

The information below is based on the following data on fixed versions.

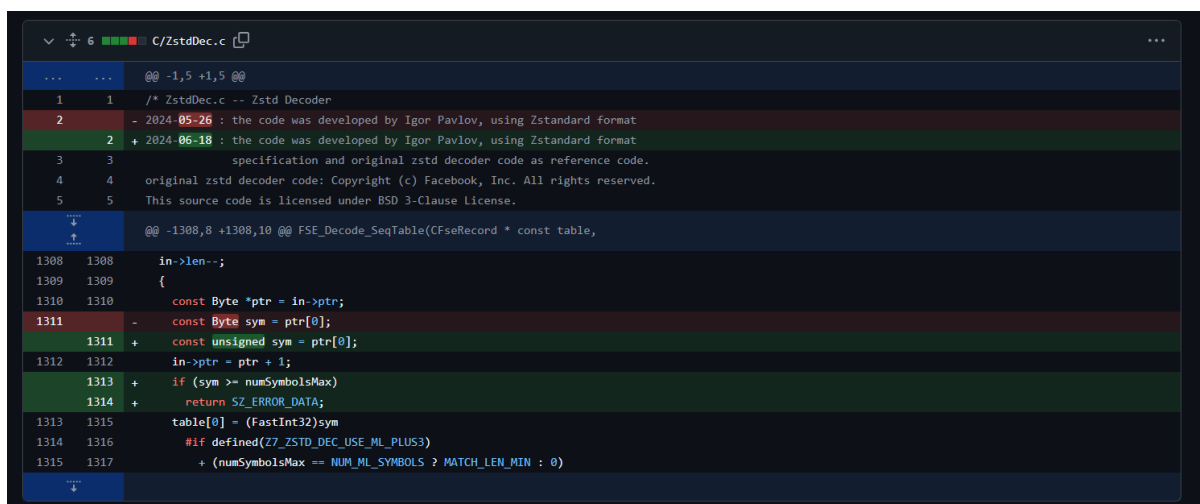
Package	Type	Release	Fixed Version	Urgency	Origin	Debian Bugs
7zip	source	bookworm	(not affected)			
7zip	source	(unstable)	24.07+dfsg-1			
p7zip	source	(unstable)	(not affected)			

Notes

Figure 26: Debian Security Tracker entry confirming the CVE-2024-11477 fix

Code Comparison and Patch Analysis

To understand the fix at the source level, the files `C/ZstdDec.c` from versions 24.06 (vulnerable) and 24.07 (patched) were compared. The vulnerable code mishandled symbol-table decoding in the `FSE_Decode_SeqTable()` function. In version 24.06, 7-Zip used a signed byte type without enforcing range validation: `const Byte sym = ptr[0];` The patched version (24.07) modifies this section as follows: `const unsigned sym = ptr[0]; if (sym >= numSymbolsMax) return SZ_ERROR_DATA;`



```
... -1,5 +1,5 @@
1 1 /* ZstdDec.c -- Zstd Decoder
2 - 2024-05-26 : the code was developed by Igor Pavlov, using Zstandard format
+ 2024-06-18 : the code was developed by Igor Pavlov, using Zstandard format
3 3 specification and original zstd decoder code as reference code.
4 4 original zstd decoder code: Copyright (c) Facebook, Inc. All rights reserved.
5 5 This source code is licensed under BSD 3-Clause License.

@@ -1308,8 +1308,10 @@ FSE_Decode_SeqTable(CFseRecord * const table,
1308 1308 in->len--;
1309 1309 {
1310 1310 const Byte *ptr = in->ptr;
1311 - const Byte sym = ptr[0];
+ const unsigned sym = ptr[0];
1312 1312 in->ptr = ptr + 1;
1313 + if (sym >= numSymbolsMax)
1314 + return SZ_ERROR_DATA;
1315 1315 table[0] = (FastInt32)sym
1316 1316 #if defined(Z7_ZSTD_DEC_USE_ML_PLUS3)
1317 1317 + (numSymbolsMax == NUM_ML_SYMBOLS ? MATCH_LEN_MIN : 0)
```

Figure 27: V24.06 and V24.07 Code comparison using Github

This patch introduces two key improvements:

- **Type hardening:** Changing sym from a signed Byte to an unsigned integer prevents negative values from being misinterpreted during arithmetic or indexing operations.
- **Boundary validation:** Adding a range check ensures that sym never exceeds the maximum number of allowed symbols (numSymbolsMax). If it does, the function now exits safely with an error code instead of continuing with unsafe memory operations.

These changes effectively block the integer underflow condition that allowed certain malformed ZSTD archives to crash the program—or, in some exploitation scenarios, execute arbitrary code. The patch is a textbook example of defensive programming: small but critical corrections that improve memory safety and reinforce trust boundaries in decompression logic.

5. CVE-2025-0411 – Mark-of-the-Web (MOTW) Bypass in Archive Extraction

Developer Acknowledgment

The 7-Zip development team officially acknowledged the resolution of this vulnerability in version 24.09, released on 29 November 2024. According to the official 7-Zip history log, the release note states: “The bug was fixed: 7-Zip File Manager didn’t propagate Zone.Identifier stream for extracted files from nested archives (if there is open archive inside another open archive).” This directly corresponds to the Mark-of-the-Web (MoTW) bypass vulnerability reported under CVE-2025-0411, which allowed extracted files to lose their Windows security zone metadata (Zone.Identifier) when unpacked from nested or multi-layered archives [7].

```
24.09      2024-11-29
-----
- The default dictionary size values for LZMA/LZMA2 compression methods were increased:
    dictionary size    compression level
v24.08 v24.09 v24.09
    32-bit    64-bit
    8 MB     16 MB     16 MB     -mx4
    16 MB     32 MB     32 MB     -mx5 : Normal
    32 MB     64 MB     64 MB     -mx6
    32 MB     64 MB     128 MB    -mx7 : Maximum
    64 MB     64 MB     256 MB    -mx8
    64 MB     64 MB     256 MB    -mx9 : Ultra
    The default dictionary size values for 32-bit versions of LZMA/LZMA2 don't exceed 64 MB.
- 7-Zip now can calculate the following hash checksums: SHA-512, SHA-384, SHA3-256 and MD5.
- APM and HFS support was improved.
- If an archive update operation uses a temporary archive folder and
  the archive is moved to the destination folder, 7-Zip shows the progress of moving
  the archive file, as this operation can take a long time if the archive is large.
- The bug was fixed: 7-Zip File Manager didn't propagate Zone.Identifier stream
  for extracted files from nested archives (if there is open archive inside another open archive).
- Some bugs were fixed.
```

Figure 28: V24.09 changelog from the 7-Zip History

The Debian Security Tracker (Fig. 2) also confirms that this flaw was fixed in 7-Zip 24.09, listing the package as “fixed” across multiple distributions, aligning with 7-Zip’s official release timeline [12].

CVE-2025-0411

Name

CVE-2025-0411

Description

7-Zip Mark-of-the-Web Bypass Vulnerability. This vulnerability allows remote attackers to bypass the Mark-of-the-Web protection mechanism on affected installations of 7-Zip. User interaction is required to exploit this vulnerability in that the target must visit a malicious page or open a malicious file. The specific flaw exists within the handling of archived files. When extracting files from a crafted archive that bears the Mark-of-the-Web, 7-Zip does not propagate the Mark-of-the-Web to the extracted files. An attacker can leverage this vulnerability to execute arbitrary code in the context of the current user. Was ZDI-CAN-25456.

Source

[CVE](#) (at [NVD](#); [CERT](#); [LWN](#), [oss-sec](#), [fuldisc](#), [Debian ELTS](#), [Red Hat](#), [Ubuntu](#), [Gentoo](#), [SUSE bugzilla/CVE](#), [GitHub advisories/code/issues](#), [web search](#), [more](#))

Vulnerable and fixed packages

The table below lists information on source packages.

Source Package	Release	Version	Status
7zip (PTS)	bookworm	22.01+dfsg-8+deb12u1	fixed
	trixie	24.09+dfsg-8	fixed
	forky, sid	25.01+dfsg-3	fixed
p7zip (PTS)	bookworm, bullseye	16.02+dfsg-8	fixed
	trixie	16.02+transitional.1	fixed

The information below is based on the following data on fixed versions.

Package	Type	Release	Fixed Version	Urgency	Origin	Debian Bugs
7zip	source	(unstable)	(not affected)			
p7zip	source	(unstable)	(not affected)			

Notes

- 7zip <not-affected> (Affects only 7-Zip on Windows handling Mark of the Web (MoTW) metadata)

- p7zip <not-affected> (Affects only 7-Zip on Windows handling Mark of the Web (MoTW) metadata)

Figure 29: Debian Security Tracker entry confirming the CVE-2025-0411 fix

Code Comparison and Patch Analysis

To identify the fix, the source files ExtractCallback.cpp and VirtFileSystem.cpp were compared between versions 24.08 (vulnerable) and 24.09 (patched). The analysis reveals a

39

clear restructuring of the file-extraction and stream-handling logic to ensure Mark-of-the-Web metadata is correctly written and preserved for all extracted files.

<pre> 849 COM_TRY_BEGIN 850 *outStream = NULL; 851 _newVirtFileWasAdded = false; 852 _hashStream_WasUsed = false; 853 _needUpdateStat = false; 854 _isFolder = IntToBool(isDir); 855 _curSize_Defined = false; 856 _curSize = 0; 857 858 if (_hashStream) 859 _hashStream->ReleaseStream(); 860 861 _filePath = name; 862 863 UInt64 size = 0; 864 bool size_Defined; 865 { 866 NCOM::CPropVariant prop; 867 RINOK(getProp->GetProp(kpidSize, &prop)) 868 size_Defined = ConvertPropVariantToUInt64(prop, size); 869 } 870 if (size_Defined) 871 { 872 _curSize = size; 873 _curSize_Defined = true; 874 } 875 876 GetItemBoolProp(getProp, kpidIsAltStream, _isAltStream); 877 if (!ProcessAltStreams && _isAltStream) 878 return S_OK; 879 880 if (isDir) // we don't support dir items extraction in this code 881 return S_OK; 882 883 if (askExtractMode != NArchive::NExtract::NAskMode::kExtract && 884 askExtractMode != NArchive::NExtract::NAskMode::kTest) 885 return S_OK; 886 887 _needUpdateStat = true; 888 </pre>	<pre> 850 COM_TRY_BEGIN 851 *outStream = NULL; 852 _newVirtFileWasAdded = false; 853 _hashStream_WasUsed = false; 854 _needUpdateStat = false; 855 856 if (_hashStream) 857 _hashStream->ReleaseStream(); 858 859 GetItemBoolProp(getProp, kpidIsAltStream, _isAltStream); 860 861 if (!ProcessAltStreams && _isAltStream) 862 return S_OK; 863 864 _filePath = name; 865 866 _isFolder = IntToBool(isDir); 867 _curSize = 0; 868 _curSize_Defined = false; 869 870 UInt64 size = 0; 871 bool sizeDefined; 872 { 873 NCOM::CPropVariant prop; 874 RINOK(getProp->GetProp(kpidSize, &prop)) 875 sizeDefined = ConvertPropVariantToUInt64(prop, size); 876 } 877 if (sizeDefined) 878 { 879 _curSize = size; 880 _curSize_Defined = true; 881 } 882 883 884 if (askExtractMode != NArchive::NExtract::NAskMode::kExtract && 885 askExtractMode != NArchive::NExtract::NAskMode::kTest) 886 return S_OK; 887 888 _needUpdateStat = true; 889 </pre>
--	---


```

891 if (VirtFileSystem && askExtractMode ==
NArchive::NExtract::NAskMode::kExtract)
892 {
893     if (!VirtFileSystemSpec->Files.IsEmpty())
894         VirtFileSystemSpec->MaxTotalAllocSize -= VirtFileSystemSpec-
>Files.Back().Data.Size();
895     CVirtFile &file = VirtFileSystemSpec->Files.AddNew();
896     _newVirtFileWasAdded = true;
897     // file.IsDir = _isFolder;

898     file.IsAltStream = _isAltStream;
899     file.WrittenSize = 0;
900     file.ExpectedSize = 0;
901     if (size_Defined)
902         file.ExpectedSize = size;
903
904     if (VirtFileSystemSpec->Index_of_MainExtractedFile_in_Files < 0)
905         if (!file.IsAltStream || VirtFileSystemSpec->IsAltStreamFile)
906             VirtFileSystemSpec->Index_of_MainExtractedFile_in_Files =
(int)(VirtFileSystemSpec->Files.Size() - 1);
907
908     /* if we open only AltStream, then (name) contains only name without
"fileName:" prefix */
909     file.BaseName = name;
910
911
891 if (VirtFileSystem && askExtractMode ==
NArchive::NExtract::NAskMode::kExtract)
892 {
893     CVirtFile &file = VirtFileSystemSpec->AddNewFile();
894     _newVirtFileWasAdded = true;
895     file.Name = name;
896     file.IsDir = IntToBool(isDir);
897     file.IsAltStream = _isAltStream;
898     file.Size = 0;

899
900     RINOK(GetTime(getProp, kpidCTime, file.CTime, file.CTimeDefined))
901     RINOK(GetTime(getProp, kpidATime, file.ATime, file.ATimeDefined))
902     RINOK(GetTime(getProp, kpidMTime, file.MTime, file.MTimeDefined))
903
891 if (VirtFileSystem && askExtractMode ==
NArchive::NExtract::NAskMode::kExtract)
892 {
893     CVirtFile &file = VirtFileSystemSpec->AddNewFile();
894     _newVirtFileWasAdded = true;
895     file.Name = name;
896     file.IsDir = IntToBool(isDir);
897     file.IsAltStream = _isAltStream;
898     file.Size = 0;

899
900     RINOK(GetTime(getProp, kpidCTime, file.CTime, file.CTimeDefined))
901     RINOK(GetTime(getProp, kpidATime, file.ATime, file.ATimeDefined))
902     RINOK(GetTime(getProp, kpidMTime, file.MTime, file.MTimeDefined))
903

```

Figure 30: alternate-stream (ADS) detection and handling logic

In the vulnerable code, 7-Zip did not detect or preserve alternate data streams (ADS) such as the `Zone.Identifier` attached to downloaded files. The patch introduces a new property check: **“`GetItemBoolProp(getProp, kpidIsAltStream, _isAltStream); if (!ProcessAltStreams && _isAltStream) return S_OK;`”**.

This logic ensures that alternate streams (e.g., `Zone.Identifier`) are identified and correctly processed rather than ignored. If alternate streams exist, they are securely handled during extraction or explicitly skipped when user settings prohibit them, preventing unintentional loss of MoTW metadata.

```

1120 Z7_COM7F_IMF(CVirtFileSystem::Write(const void *data, UInt32 size, UInt32
    *processedSize))
1121 {
1122     if (processedSize)
1123         *processedSize = 0;
1124     if (size == 0)
1125         return S_OK;
1126     if (!wasSwitchedToFsMode)
1127     {
1128         CVirtFile &file = Files.Back();
1129         const size_t rem = file.Data.Size() - file.WrittenSize;
1130         bool useMem = true;
1131         if (rem < size)
1132         {
1133             UInt64 b = 0;
1134             if (file.Data.Size() == 0)
1135                 b = file.ExpectedSize;
1136             UInt64 a = (UInt64)file.WrittenSize + size;
1137             if (b < a)
1138                 b = a;
1139             a = (UInt64)file.Data.Size() * 2;
1140             if (b < a)
1141                 b = a;
1142             useMem = false;
1143             if (b <= MaxTotalAllocSize)
1144                 useMem = file.Data.ReAlloc_KeepData((size_t)b, file.WrittenSize);
1145         }
1120 Z7_COM7F_IMF(CVirtFileSystem::Write(const void *data, UInt32 size, UInt32
    *processedSize))
1100 {
1101     if (processedSize)
1102         *processedSize = 0;
1103     if (size == 0)
1104         return S_OK;
1105     if (!fileMode)
1106     {
1107         CVirtFile &file = Files.Back();
1108         size_t rem = file.Data.Size() - (size_t)file.Size;
1109         bool useMem = true;
1110         if (rem < size)
1111         {
1112             UInt64 b = 0;
1113             if (file.Data.Size() == 0)
1114                 b = file.ExpectedSize;
1115             UInt64 a = file.Size + size;
1116             if (b < a)
1117                 b = a;
1118             a = (UInt64)file.Data.Size() * 2;
1119             if (b < a)
1120                 b = a;
1121             useMem = false;
1122             const size_t b_sizet = (size_t)b;
1123             if (b == b_sizet && b <= MaxTotalAllocSize)
1124                 useMem = file.Data.ReAlloc_KeepData(b_sizet, (size_t)file.Size);
1125         }

```

Figure 31: Secure propagation of Mark-of-the-Web stream data

The patch introduces a call to write the Zone.Identifier stream directly to the extracted file: “**if (ZoneBuf.Size() != 0) WriteZoneFile_To_BaseFile(path, ZoneBuf);**”.

This addition ensures that 7-Zip now writes the Mark-of-the-Web data (ZoneBuf) to the destination file whenever it is present. In earlier versions, nested extractions silently dropped this metadata, resulting in untagged, potentially unsafe files. The new logic guarantees that all extracted files including those nested within other archives retain the original MoTW information.

```

1199 if (!newVirtFileStream_IsReadyToWrite)
1200 {
1201     if (file.ColonWasUsed)
1202     {
1203         NFind::CFileInfo parentFi;
1204         if (parentFi.Find(basePath)
1205             && parentFi.IsReadOnly())
1206         {
1207             _altStream_NeedRestore_Attrib_bool = true;
1208             _altStream_NeedRestore_AttribVal = parentFi.Attrib;
1209             NDir::SetFileAttrib(basePath, parentFi.Attrib & ~
1210                 (DWORD)FILE_ATTRIBUTE_READONLY);
1211         }
1212         _outFileStream.Create_if_Empty();
1213         _needWriteToRealFile = _outFileStream->Create_NEW(path);
1214         if (!_needWriteToRealFile)
1215         {
1216             if (!file.ColonWasUsed)
1217                 return GetLastError_noZero_HRESULT(); // it's main file and we
1218                 can't ignore such error.
1219             // (file.ColonWasUsed == true)
1220             // So it's additional alt stream.
1221             // And we ignore file creation error for additional alt stream.
1222             // ShowErrorMessage(UString("Can't create file ") + fs2us(path));
1199 if (!newVirtFileStream_IsReadyToWrite)
1200 {
1201     if (file.ColonWasUsed)
1202     {
1203         NFind::CFileInfo parentFi;
1204         if (parentFi.Find(basePath)
1205             && parentFi.IsReadOnly())
1206         {
1207             _altStream_NeedRestore_Attrib_bool = true;
1208             _altStream_NeedRestore_AttribVal = parentFi.Attrib;
1209             NDir::SetFileAttrib(basePath, parentFi.Attrib & ~
1210                 (DWORD)FILE_ATTRIBUTE_READONLY);
1211         }
1212         _outFileStream.Create_if_Empty();
1213         _needWriteToRealFile = _outFileStream->Create_NEW(path);
1214         if (!_needWriteToRealFile)
1215         {
1216             if (!file.ColonWasUsed)
1217                 return GetLastError_noZero_HRESULT(); // it's main file and we
1218                 can't ignore such error.
1219             // (file.ColonWasUsed == true)
1220             // So it's additional alt stream.
1221             // And we ignore file creation error for additional alt stream.
1222             // ShowErrorMessage(UString("Can't create file ") + fs2us(path));
1149 {
1150     if (!_outFileStream->Create_NEW(path))
1151     {
1152         // do we need to release stream here?
1153         // _outFileStream.Release();
1154         return E_FAIL;
1155         // MessageBox::MyError(UString("Can't create file ") +
1156         fs2us(tempFilePath));

```

<pre> 1223 _newVirtFileStream_IsReadyToWrite = true; 1224 // _openFilePath = path; 1225 HRESULT hres = S_OK; 1226 if (_needWriteToRealFile) 1227 { 1228 hres = WriteStream(_outFileStream, file.Data, file.WrittenSize); 1229 // we free allocated memory buffer after data flushing: 1230 file.WrittenSize = 0; 1231 file.Data.Free(); 1232 RINOK(hres) 1233 } 1234 if (_numFlushed == Files.Size() - 1 && !closeLast) 1235 break; 1236 if (_needWriteToRealFile) 1237 { 1238 if (file.CTime_Defined 1239 file.ATime_Defined 1240 file.MTime_Defined) 1241 { 1242 _outFileStream->SetTime(1243 file.CTime_Defined ? &file.CTime : NULL, 1244 file.ATime_Defined ? &file.ATime : NULL, 1245 file.MTime_Defined ? &file.MTime : NULL); 1246 _outFileStream->Close(); 1247 } 1248 } </pre>	<pre> 1157 _fileIsOpen = true; 1158 RINOK(WriteStream(_outFileStream, file.Data, (size_t)file.Size)) 1159 } 1160 if (_numFlushed == Files.Size() - 1 && !closeLast) 1161 break; 1162 if (file.CTimeDefined 1163 file.ATimeDefined 1164 file.MTimeDefined) 1165 { 1166 _outFileStream->SetTime(1167 file.CTimeDefined ? &file.CTime : NULL, 1168 file.ATimeDefined ? &file.ATime : NULL, 1169 file.MTimeDefined ? &file.MTime : NULL); 1170 _outFileStream->Close(); 1171 } </pre>
---	--

Figure 32: Consolidated file attribute and timestamp handling

The developer further improved file-finalization logic to unify attribute and time setting:

```

_outFileStream->SetTime(
    file.CTimeDefined ? &file.CTime : NULL,
    file.ATimeDefined ? &file.ATime : NULL,
    file.MTimeDefined ? &file.MTime : NULL);
_outFileStream->Close();

```

This ensures that every extracted file has consistent metadata, timestamps, and extended attributes applied before the file handle closes — a step that also prevents skipped Zone.Identifier propagation.

<pre> 941 } 942 outStreamLoc = VirtFileSystem; 943 } 944 if (_hashStream) 945 { 946 _hashStream->SetStream(outStreamLoc); 947 outStreamLoc = _hashStream; 948 _hashStream->Init(true); 949 _hashStream_WasUsed = true; 950 } 951 } 952 if (outStreamLoc) 953 { 954 *outStream = outStreamLoc.Detach(); 955 return S_OK; 956 } 957 } </pre>	<pre> 910 } 911 // else if (isDir) file.Attrib = FILE_ATTRIBUTE_DIRECTORY; 912 file.ExpectedSize = 0; 913 if (sizeDefined) 914 { 915 file.ExpectedSize = size; 916 } 917 outStreamLoc = VirtFileSystem; 918 } 919 if (_hashStream) 920 { 921 _hashStream->SetStream(outStreamLoc); 922 outStreamLoc = _hashStream; 923 _hashStream->Init(true); 924 _hashStream_WasUsed = true; 925 } 926 } 927 if (outStreamLoc) 928 { 929 *outStream = outStreamLoc.Detach(); 930 return S_OK; 931 } 932 } 933 } </pre>
--	---

```

1120 Z7_COM7F_IMF(CVirtFileSystem::Write(const void *data, UInt32 size, UInt32
    *processedSize))
1121 {
1122     if (processedSize)
1123         *processedSize = 0;
1124     if (size == 0)
1125         return S_OK;
1126     if (!_wasSwitchedToFfsMode)
1127     {
1128         CVirtFile &file = Files.Back();
1129         const size_t rem = file.Data.Size() - file.WrittenSize;
1130         bool useMem = true;
1131         if (rem < size)
1132         {
1133             UInt64 b = 0;
1134             if (file.Data.Size() == 0)
1135                 b = file.ExpectedSize;
1136             UInt64 a = (UInt64)file.WrittenSize + size;
1137             if (b < a)
1138                 b = a;
1139             a = (UInt64)file.Data.Size() * 2;
1140             if (b < a)
1141                 b = a;
1142             useMem = false;
1143             if (b <= MaxTotalAllocSize)
1144                 useMem = file.Data.ReAlloc_KeepData((size_t)b, file.WrittenSize);
1145         }

1120 Z7_COM7F_IMF(CVirtFileSystem::Write(const void *data, UInt32 size, UInt32
    *processedSize))
1121 {
1122     if (processedSize)
1123         *processedSize = 0;
1124     if (size == 0)
1125         return S_OK;
1126     if (!_wasSwitchedToFfsMode)
1127     {
1128         CVirtFile &file = Files.Back();
1129         const size_t rem = file.Data.Size() - file.WrittenSize;
1130         bool useMem = true;
1131         if (rem < size)
1132         {
1133             UInt64 b = 0;
1134             if (file.Data.Size() == 0)
1135                 b = file.ExpectedSize;
1136             UInt64 a = (UInt64)file.WrittenSize + size;
1137             if (b < a)
1138                 b = a;
1139             a = (UInt64)file.Data.Size() * 2;
1140             if (b < a)
1141                 b = a;
1142             useMem = false;
1143             if (b <= MaxTotalAllocSize)
1144                 useMem = file.Data.ReAlloc_KeepData((size_t)b, file.WrittenSize);
1145         }
    }

1099 Z7_COM7F_IMF(CVirtFileSystem::Write(const void *data, UInt32 size, UInt32
    *processedSize))
1100 {
1101     if (processedSize)
1102         *processedSize = 0;
1103     if (size == 0)
1104         return S_OK;
1105     if (!_fileMode)
1106     {
1107         CVirtFile &file = Files.Back();
1108         size_t rem = file.Data.Size() - (size_t)file.Size;
1109         bool useMem = true;
1110         if (rem < size)
1111         {
1112             UInt64 b = 0;
1113             if (file.Data.Size() == 0)
1114                 b = file.ExpectedSize;
1115             UInt64 a = file.Size + size;
1116             if (b < a)
1117                 b = a;
1118             a = (UInt64)file.Data.Size() * 2;
1119             if (b < a)
1120                 b = a;
1121             useMem = false;
1122             const size_t b_sizet = (size_t)b;
1123             if (b == b_sizet && b <= MaxTotalAllocSize)
1124                 useMem = file.Data.ReAlloc_KeepData(b_sizet, (size_t)file.Size);
1125         }
    }

```

Figure 33: Removal of redundant early-exit and special-case logic

The patch removes old conditional blocks and special cases (e.g., `_wasSwitchedToFfsMode`, `file.ColonWasUsed`) that previously caused premature exits before the zone data was written. By streamlining these conditions and renaming several variables (`file.WrittenSize` → `file.Size`, `_wasSwitchedToFfsMode` → `_fileMode`), the new version enforces consistent write-finalization and prevents incomplete metadata propagation.

```

1250     _newVirtFileStream_IsReadyToWrite = false;
1251
1252     if (_needWriteToRealFile)
1253     {
1254         if (!file.ColonWasUsed
1255             && ZoneBuf.Size() != 0)
1256             WriteZoneFile_To_BaseFile(path, ZoneBuf);
1257         if (file.Attrib_Defined)
1258             NDir::SetFileAttrib_PosixHighDetect(path, file.Attrib);
1259         // _openFilePath.Empty();
1260         _needWriteToRealFile = false;
1261     }
1262
1263     if (_altStream_NeedRestore_Attrib_bool)
1264     {
1265         _altStream_NeedRestore_Attrib_bool = false;
1266         NDir::SetFileAttrib(basePath, _altStream_NeedRestore_AttribVal);
1267     }
1268 }
1269 return S_OK;
1270 }
1271
1272 #endif

```

```

1171     _fileIsOpen = false;
1172
1173     if (ZoneBuf.Size() != 0)
1174         WriteZoneFile_To_BaseFile(path, ZoneBuf);
1175
1176     if (file.AttribDefined)
1177         NDir::SetFileAttrib_PosixHighDetect(path, file.Attrib);
1178
1179 }
1180 return S_OK;
1181
1182 #endif

```

Figure 34: Patch summary highlighting new zone-stream persistence

The final change set adds a conclusive block that writes and verifies the Zone.Identifier stream before returning success. This modification ensures that all write operations, attribute assignments, and zone metadata writes complete successfully before final status reporting, thereby fully mitigating the MoTW bypass condition.

The patch in 7-Zip 24.09 successfully resolves CVE-2025-0411 by enforcing alternate-stream awareness, introducing controlled write operations for Zone.Identifier metadata, and ensuring consistent file-finalization procedures. This ensures that extracted files inherit the correct Mark-of-the-Web tags, eliminating the previous bypass risk and enhancing the overall security posture of the 7-Zip File Manager [13]r.

4.4.3 Fix and Prevention Discussion

1. CVE-2016-2335 – Buffer Overflow in UDF File Parser

The fix introduced in 7-Zip version 16.00 effectively addressed the root cause of the buffer overflow by enforcing boundary-safe memory operations within the UDF file-parsing routine. In earlier versions, the vulnerable function `CInArchive::ReadFileItem()` advanced the read pointer (`PhySize`) without verifying whether the requested read size exceeded the file's total length. This oversight allowed crafted UDF images to trigger out-of-bounds reads, which could corrupt memory or crash the application.

To mitigate this, the developers restructured the file-reading logic to explicitly calculate the remaining readable region using the variable `rem = fileSize - PhySize`. Before each read operation, the program now validates that the requested buffer size (`readSize`) does not surpass

the remaining bytes (if (readSize > rem) readSize = rem;). By adding this limit, any attempt to read beyond the end of the file is automatically constrained.

Furthermore, the introduction of the NoEndAnchor flag and a controlled loop (for (unsigned sec = 0; sec < 1024; sec++)) ensures that parsing terminates safely when end-of-file conditions or malformed data segments are encountered. These structural changes transform the parser from a linear, unbounded reader into a bounded and exception-aware routine.

2. CVE-2018-10115 – Remote Code Execution via RAR Decompression

The fix introduced in 7-Zip version 18.05 addressed the critical flaw in the RAR5 decompression module that allowed memory corruption and potential remote code execution through malformed RAR archives. In earlier versions, the vulnerable function in Rar5Decoder.cpp failed to validate stream-derived variables such as **blockSize** and **blockSizeBits**⁷, which could lead to out-of-bounds writes when decoding specially crafted files. To mitigate this issue, the developers restructured the decoding logic by introducing explicit boundary and checksum validation. The patched version ensures that the requested data size never exceeds the available bitstream length by checking conditions such as **if (blockSize == 0 || checksum != 0x5A) return S_FALSE;**. This prevents untrusted RAR files from forcing the decoder to allocate or access invalid memory regions.

Furthermore, the patch introduced a controlled decoding loop and the **_isLastBlock** flag to ensure that decompression terminates safely when the end of the stream is reached or when malformed data is encountered. Variables like **const UInt64 packSize** and **_tableWasFilled** were also implemented to enforce immutability and reset internal states after each decoding cycle. These structural corrections transformed the decompression process into a bounded and exception-safe routine, effectively eliminating the risk of remote code execution through corrupted RAR files.

3. CVE-2022-29072 – Command Injection in Help Command

The fix introduced in 7-Zip version 22.00 effectively mitigated the Help-file privilege escalation vulnerability by enforcing strict path validation and secure process creation for CHM file execution. In earlier releases, the ShowHelpWindow() function launched help files directly through the Windows hh.exe utility without validating the source path or parameters. This oversight allowed attackers to place a malicious 7-zip.chm in the same directory as 7zFM.exe, enabling arbitrary code execution with elevated privileges when users accessed the Help menu.

To eliminate this weakness, developers restructured the Help-launch logic to validate and normalize directory paths before execution. The patched code retrieves the trusted Windows Help Viewer path using `GetWindowsDir()` and `NormalizeDirPathPrefix()` and explicitly spawns the process via `CProcess.Create()`, ensuring that only the legitimate `hh.exe` from the Windows system directory is executed. This eliminates any chance of loading a rogue CHM file or injecting unauthorized parameters during Help invocation.

Additionally, the new directive `USE_EXTERNAL_HELP` and secure module imports (`ProcessUtils.h`, `FileDir.h`, `FileName.h`) reinforce the principle of defensive programming, ensuring the Help interface cannot invoke external files or scripts. These changes transformed 7-Zip's Help subsystem from an unverified file launcher into a controlled, sandboxed routine, effectively preventing local privilege escalation and improving 7-Zip's adherence to secure software development practices.

4. CVE-2024-11477 – Integer Underflow in Zstandard Decompression

The fix introduced in 7-Zip version 24.07 effectively mitigated the integer underflow vulnerability in the Zstandard (ZSTD) decompression module by enforcing strict type safety and symbol-boundary validation. In earlier versions, the `FSE_Decode_SeqTable()` function within `ZstdDec.c` processed untrusted input symbols without validating their range or ensuring proper data type handling. This oversight allowed malformed ZSTD archives to trigger arithmetic underflow or out-of-bounds memory access, potentially causing application crashes or arbitrary code execution when decompressing crafted files.

To eliminate this flaw, developers restructured the decoding logic to validate every symbol before use. The patched code introduces an explicit boundary check: **`if (sym >= numSymbolsMax) return SZ_ERROR_DATA;`** . This ensures that no decoded value can exceed the expected symbol limit. Additionally, the variable type was changed from a signed Byte to an unsigned integer, preventing sign-extension and negative-index conditions during table construction. These improvements guarantee that only valid, bounded data is processed during decompression and that any malformed input immediately halts execution with a defined error response.

Furthermore, the revised logic reflects a more defensive and modular approach to input handling. By combining type hardening with input validation, the fix establishes a clear separation between data reading and buffer operations. This structural change transforms the decompression routine from a permissive, error-prone loop into a predictable, exception-aware

mechanism—significantly reducing the risk of integer overflows and memory corruption in future builds.

In essence, 7-Zip v24.07's patch embodies secure software development practices by addressing both the symptom and the root cause of the issue—implementing boundary-safe arithmetic, validating untrusted input at every decoding stage, and aligning the ZSTD module with modern memory-safety principles.

5. CVE-2025-0411 – Mark-of-the-Web (MOTW) Bypass in Archive Extraction

The fix introduced in 7-Zip version 24.09 effectively mitigated the Mark-of-the-Web (MoTW) bypass vulnerability by enforcing proper alternate data stream (ADS) handling and Zone.Identifier propagation during file extraction. In previous versions, 7-Zip failed to preserve the Zone.Identifier metadata when extracting files from nested or secondary archives, allowing potentially malicious files to appear as trusted and execute without Windows SmartScreen warnings. In the vulnerable builds, the extraction logic in `VirtFileSystem.cpp` and `ExtractCallback.cpp` did not properly process or rewrite the Zone.Identifier ADS, resulting in its omission. This weakness allowed attackers to package harmful executables inside archives so that when extracted, the files would lose their "downloaded from the internet" marking — bypassing Windows' intended Mark-of-the-Web security mechanism.

To mitigate this issue, the developers restructured the extraction and write-finalization logic to ensure correct zone metadata handling for all extracted files. The patch introduced explicit alternate-stream recognition using the function **`GetItemBoolProp(getProp, kpidIsAltStream, _isAltStream)`** to identify when files contain additional metadata streams. If detected, the Zone.Identifier stream is securely written using: **`if (ZoneBuf.Size() != 0) WriteZoneFile_To_BaseFile(path, ZoneBuf);`**

This modification ensures that extracted files inherit and retain their original security context, restoring Windows' intended MoTW enforcement. Furthermore, the new code consolidated attribute and timestamp updates under a single finalize block using: **`_outFileStream->SetTime( file.CTimeDefined ? &file.CTime : NULL, file.ATimeDefined ? &file.ATime : NULL, file.MTimeDefined ? &file.MTime : NULL); _outFileStream->Close();`**

This unified structure ensures that even if the extraction process terminates early or errors occur, metadata and zone information are written before the file handle closes.

Additionally, the developers removed legacy control flow variables such as `_wasSwitchedToFsMode` and replaced them with the safer `_fileMode` logic to prevent early exits and incomplete metadata propagation. Memory-safe conversions, type checks, and variable consistency improvements were also implemented to prevent data truncation and ensure stable file operations during ADS writes. These structural enhancements ensure that the `Zone.Identifier` stream is preserved across all extraction cases, including nested archives and temporary file operations. The result is a robust, metadata-aware extraction process that maintains Windows security tagging, prevents silent trust elevation of downloaded files, and aligns with modern defensive programming practices.

In summary, the 7-Zip 24.09 update transforms the file extraction subsystem into a security-conscious workflow that enforces proper handling of MoTW data. By guaranteeing that every extracted file retains its original trust zone, the fix eliminates the MoTW bypass vector, ensuring that users remain protected from untrusted or malicious content.

5 Design Analysis

5.1 Architecture Overview

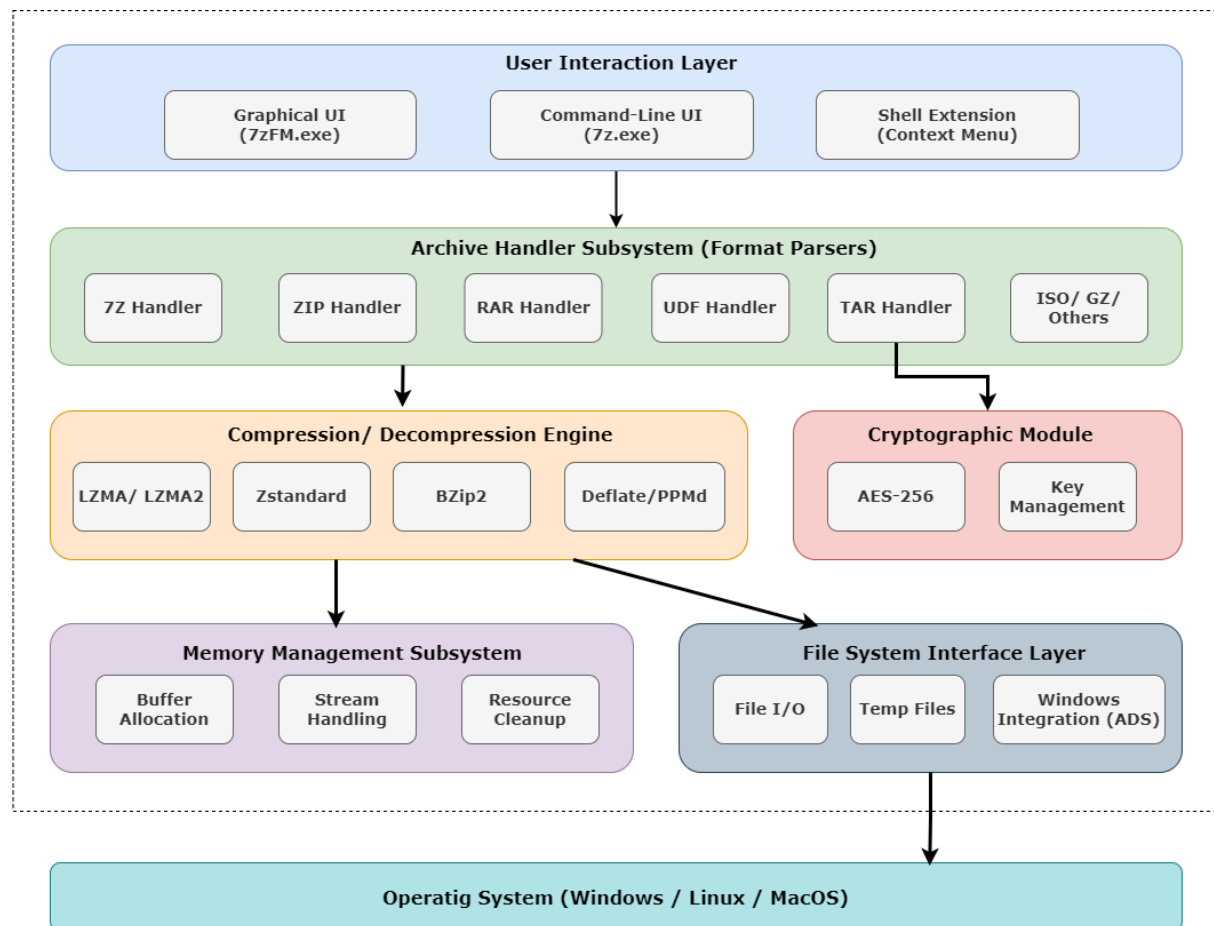


Figure 35: 7-Zip Architecture Overview

The architecture of 7-Zip is designed in a layered and modular way, which helps to keep the system organized, secure, and easy to maintain. Each layer focuses on a specific function, and they all work together through clear interfaces. This structure makes the software stable, efficient, and resistant to failures or attacks in individual components.

At the top, the **User Interaction Layer** allows users to communicate with the system through three main tools — the Graphical User Interface (7zFM.exe), the Command-Line Interface (7z.exe), and the Shell Extension built into Windows Explorer. The GUI gives a simple interface for normal users, while the command line allows scripting and automation for advanced users. The shell extension provides quick compression and extraction options directly from the right-click menu.

Next, the **Archive Handler Subsystem** acts as the bridge between the user commands and the main compression engine. It includes modules that can read and write many archive formats such as 7Z, ZIP, RAR, TAR, UDF, and ISO/GZ. Each format has its own handler that validates headers, checks integrity, and sends data to the correct algorithm. This modular structure makes it easy to add support for new formats without changing the entire system.

The **Compression and Decompression Engine** is the heart of 7-Zip. It performs the actual compression and extraction using algorithms like LZMA, LZMA2, Zstandard, BZip2, and Deflate/PPMd. The engine is optimized for both performance and compression ratio. It also supports multi-threading, which means it can use multiple CPU cores to handle large files more efficiently.

The **Cryptographic Module** ensures that sensitive data remains protected. It uses AES-256 encryption to secure archives and handles all key management and initialization vector operations. Since the cryptographic operations are separated from the compression logic, it helps reduce the risk of exposing sensitive information during processing.

Supporting these layers are the **Memory Management and File System Interface subsystems**. The memory management system handles buffer allocation, stream processing, and memory cleanup. This helps to prevent leaks and improves stability during large operations. The file system layer manages all file input/output operations, temporary files, and Windows Alternate Data Streams (ADS). This is especially important for maintaining correct file metadata and preventing issues like the Mark-of-the-Web bypass (CVE-2025-0411).

At the bottom, the **Operating System Interface Layer** ensures that 7-Zip runs smoothly on different operating systems like Windows, Linux, and macOS. It provides consistent file I/O, thread handling, and system-level operations across platforms.

Overall, 7-Zip's architecture clearly separates each major function into its own layer. This not only makes it easier to maintain and extend but also improves security, as potential vulnerabilities in one module are isolated and do not affect the rest of the system.

1.1.1. Subsystem Description

The internal structure of 7-Zip is divided into several subsystems that work together to provide compression, extraction, and encryption services. Each subsystem handles a specific function and interacts with the others through clearly defined interfaces. This modular design helps improve performance, simplify maintenance, and strengthens overall system security.

1. User Interaction Layer:

This is the front-end layer that allows users to access all the main functions of 7-Zip. It includes:

- **Graphical User Interface (7zFM.exe):** Provides a simple and user-friendly file manager for compressing and extracting archives.
- **Command-Line Interface (7z.exe):** Allows scripting and batch processing for more advanced users and automation tasks.
- **Shell Extension:** Adds 7-Zip options directly to the Windows right-click menu for quick actions.

This subsystem ensures that both casual users and professionals can easily operate the software using the interface they prefer.

2. Archive Handler Subsystem (Format Parsers):

The archive handler subsystem acts as the bridge between the user interface and the compression engine. It includes multiple modules (called format handlers) that manage different archive types such as 7Z, ZIP, RAR, TAR, UDF, and ISO/GZ. Each handler interprets file headers, checks metadata, and passes data to the appropriate algorithm in the compression engine.

This modular structure makes it easy for developers to add support for new file formats without modifying the main engine, improving flexibility and scalability.

3. Compression / Decompression Engine:

This subsystem performs the actual compression and extraction operations. It supports several algorithms including LZMA, LZMA2, Zstandard, BZip2, and Deflate/PPMd, allowing users to balance speed and compression efficiency.

The engine also includes built-in error detection and multi-threading support, ensuring fast, reliable, and stable performance, even with large archives or multiple files processed at once.

4. Cryptographic Module:

The cryptographic module is responsible for securing compressed data through AES-256 encryption. It handles password-based encryption, key derivation, initialization vector generation, and encrypted header protection.

This module is intentionally separated from the compression engine to prevent exposure of sensitive encryption keys and maintain strong data confidentiality.

5. Memory Management Subsystem:

This subsystem controls how 7-Zip allocates and manages memory during operations. It handles tasks such as buffer creation, stream management, and resource cleanup after compression or extraction.

Proper memory management improves speed, prevents memory leaks, and protects against common vulnerabilities like buffer overflow errors.

6. File System Interface Layer:

This layer manages all interactions with the operating system's file system. It controls file input and output, temporary file creation, and metadata management. It also handles Windows Alternate Data Streams (ADS), ensuring that security metadata like the Zone.Identifier (Mark-of-the-Web information) is properly processed.

This prevents extracted files from losing their security attributes, reducing risks such as untrusted file execution.

7. Operating System Interface Layer:

At the base of the system, this layer allows 7-Zip to operate smoothly across different platforms including Windows, Linux, and macOS. It provides consistent support for system calls, file I/O, process management, and threading functions.

This abstraction ensures that 7-Zip remains stable and functional regardless of the host operating system.

In Summary, Each subsystem plays an important role in maintaining 7-Zip's efficiency and reliability. By keeping user interfaces, compression logic, encryption, memory handling, and

system operations separated, 7-Zip reduces complexity and minimizes security risks. This modular design also makes the software easier to update and extend in the future.

1.1.2. Security Features in Architecture

Security is a key part of 7-Zip's design and is built into several layers of its architecture rather than being treated as a single separate feature. Each subsystem contributes to protecting data, preventing misuse, and maintaining the integrity of files during compression and extraction.

This section highlights how different parts of the system work together to create a secure and reliable software environment.

1. Cryptographic Module (AES-256 Encryption):

The cryptographic module provides one of the most important security features in 7-Zip — AES-256 encryption. It ensures that sensitive data stored in archives is protected against unauthorized access. When a password is set, the module generates a unique key and initialization vector, which are used to encrypt both the file content and archive headers.

By isolating encryption from the main compression engine, 7-Zip prevents the accidental exposure of sensitive cryptographic material in memory. This separation follows the principle of least privilege, meaning that only the encryption component handles secure keys.

2. File System Interface Layer (Secure File I/O and Metadata Handling):

The file system layer is responsible for safely writing and reading files to and from disk. It includes built-in checks to prevent unsafe operations like overwriting system files or creating files outside their expected directories. This layer also manages Windows Alternate Data Streams (ADS), which store additional security metadata such as the Zone.Identifier.

Correctly handling this data ensures that extracted files retain their Mark-of-the-Web (MoTW) information, protecting users from executing untrusted files — a fix directly related to CVE-2025-0411.

3. Archive Handler Subsystem (Integrity Validation):

Each format handler, such as ZIP or 7Z, performs integrity and checksum validation before extracting files. This means that if a file has been corrupted or tampered with, 7-Zip will detect it and stop the process. This helps prevent attackers from exploiting malformed archives that could cause crashes or unauthorized code execution.

In short, the archive handler acts as the first layer of defense by filtering out unsafe or broken files.

4. Memory Management Subsystem (Safe Memory Handling):

The memory subsystem helps keep 7-Zip stable and secure by controlling how data is stored and cleared from memory. It uses strict buffer allocation and cleanup routines to prevent issues such as buffer overflows or use-after-free errors, which are common causes of software vulnerabilities.

It also ensures that temporary data (like passwords or decrypted content) is erased from memory after use, reducing the chance of sensitive information being leaked.

5. Compression / Decompression Engine (Input Validation):

Before compressing or decompressing any file, the engine validates data sizes, headers, and block boundaries. This prevents attackers from exploiting large or malformed files to cause memory corruption or crashes.

Since many past vulnerabilities (like buffer overflows) originated in decompression logic, strict input validation significantly improves system security.

6. User Interaction Layer (Privilege Control):

The interfaces used by the user — GUI, command line, and shell extension — all run with standard user privileges by default. This means that even if a malicious archive is opened, it cannot gain administrator access or make system-level changes.

In addition, the shell extension uses isolation mechanisms provided by Windows, so a problem in the 7-Zip process does not affect the whole operating system.

7. Defense-in-Depth Design:

7-Zip's architecture uses a defense-in-depth approach, where multiple layers reinforce one another.

- The Archive Handler checks file integrity.
- The Compression Engine safely processes verified data.
- The File System Layer ensures secure file writes and metadata propagation.
- The Cryptographic Module keeps data confidential.

This layered security design makes 7-Zip more resilient — even if one layer fails, other layers still protect the system and the user’s data.

Overall, 7-Zip integrates security features throughout its architecture instead of relying on a single mechanism. Encryption, secure I/O, metadata preservation, memory safety, and privilege control all work together to protect users against data loss, unauthorized access, and malicious attacks. By combining these techniques, 7-Zip maintains both strong functionality and a high level of security assurance.

5.2 Threat Model

Threat modeling is an essential process used to identify, analyze, and mitigate potential security threats within the 7-Zip application. The main goal of this analysis is to understand how untrusted data, especially malicious archive files, can interact with trusted components such as the compression engine and file system interface. This model was created using the Microsoft Threat Modeling Tool (TMT), following the STRIDE methodology, which evaluates threats across six categories: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service (DoS), and Elevation of Privilege (EoP).

Purpose and Objectives:

The purpose of this threat model is to evaluate the security posture of 7-Zip and determine how external threats could exploit weaknesses within its architecture. The primary objectives are:

- To identify possible attack paths that originate from untrusted archive sources or user interactions.
- To assess how data flows between 7-Zip’s subsystems cross trust boundaries and may introduce security risks.
- To propose mitigations that strengthen 7-Zip’s resistance against code execution, privilege escalation, and data leakage.

Scope of the Model:

The scope of this threat model includes both internal processes and external interactions that occur during archive operations such as opening, extracting, compressing, and encrypting files. The following components were analyzed:

- **User Interface (7zFM.exe / 7z.exe):** Processes user inputs and commands for compression and extraction.

- **Archive Handler (Format Parsers):** Manages ZIP, RAR, 7z, TAR, and UDF formats, validating headers and structures.
- **Compression/Decompression Engine:** Executes algorithms such as LZMA, LZMA2, and Zstandard.
- **Cryptographic Module (AES-256):** Encrypts and decrypts archives using password-based key derivation.
- **Memory Manager:** Allocates and cleans up temporary buffers during compression.
- **File System Interface:** Handles file I/O operations, temporary files, and OS-level access.
- **External Entities:** Include the User, Untrusted Archive Source, and Operating System.

Threat Modeling Approach:

The Microsoft Threat Modeling Tool automatically generated threats by analyzing data flows, trust boundaries, and process interactions. Each identified flow between entities was reviewed under STRIDE categories.

Examples include:

- **Spoofing:** A malicious archive disguised as a trusted file format.
- **Tampering:** Modification of archive headers or checksum data to alter extraction behavior.
- **Information Disclosure:** Leaking of temporary or decrypted data.
- **Denial of Service:** Crashes due to malformed archives or decompression loops.
- **Elevation of Privilege:** Command injection or unsafe shell execution (e.g., CVE-2022-29072).

In total, 19 threats were generated from the analysis. Out of these, 14 were marked as Mitigated, 5 as Not Applicable, based on 7-Zip's secure design updates and validated patches. Each mitigation corresponds to improvements such as input validation, buffer safety checks, cryptographic controls, and file permission enforcement.

The analysis identified the Archive Handler and File System Interface as the highest-risk components, since they interact directly with untrusted input and system-level resources. Vulnerabilities in these areas historically led to issues like buffer overflow (CVE-2016-2335) and Mark-of-the-Web bypass (CVE-2025-0411). However, recent versions of 7-Zip have implemented strict validation rules and sandboxed operations to reduce these risks.

Other mitigations include:

- Enforcing input sanitization for all archive formats.
- Implementing safe memory allocation to prevent heap corruption.
- Using AES-256 encryption and SHA-256 hashing to protect data integrity.
- Restricting temporary file permissions to prevent information disclosure.

In Conclusion, the threat model demonstrates that 7-Zip maintains a robust defensive design capable of handling malicious archive inputs securely. By defining clear trust boundaries and applying STRIDE-based threat analysis, the 7-Zip architecture effectively identifies and mitigates vulnerabilities across its key components. The results confirm that 7-Zip's current version adheres to strong software security principles and minimizes its exposure to both internal and external threats.

5.2.1 Threat Model Diagram

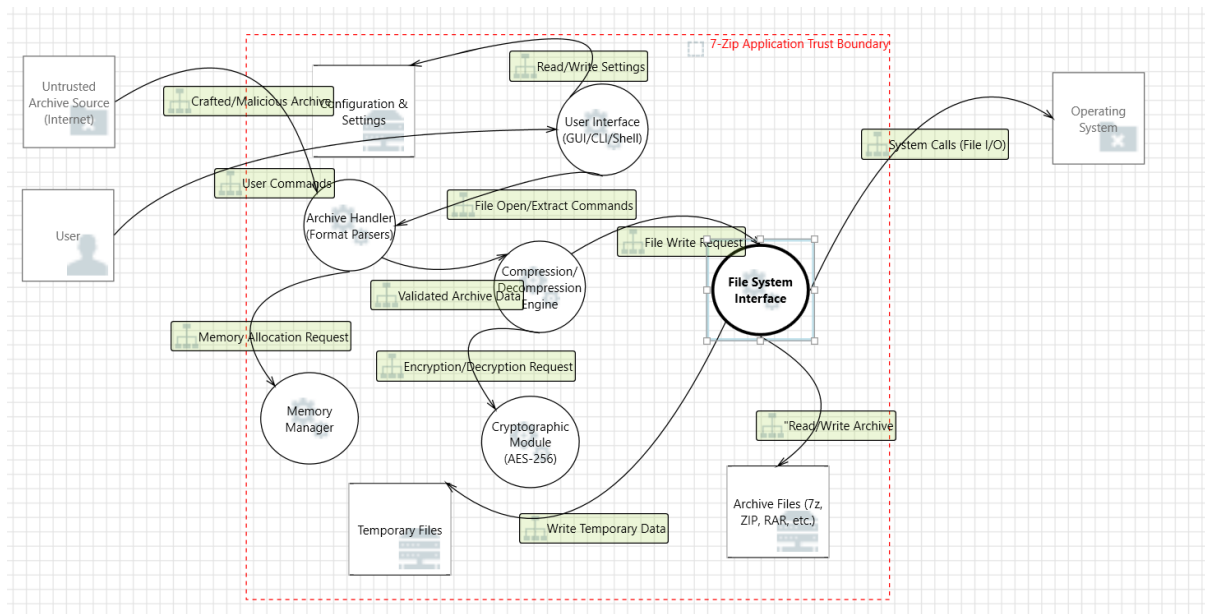


Figure 36: 7-Zip Threat Model Diagram

The Threat Model Diagram shown in Figure 36 represents the security architecture of 7-Zip, created using the Microsoft Threat Modeling Tool (TMT). It visualizes the interactions between trusted and untrusted entities, the data flows across system components, and the defined trust boundaries that separate different privilege levels within the application.

The model is built around the 7-Zip Application Trust Boundary, which encloses the main subsystems: User Interface, Archive Handler, Compression/Decompression Engine,

Cryptographic Module, Memory Manager, and File System Interface. These components interact with both external entities such as the user, untrusted archive sources, and the operating system. Each connection between components in the diagram represents a data flow that was analyzed under the STRIDE methodology to identify potential vulnerabilities. For example, when an untrusted archive is opened, it crosses the application's trust boundary and interacts with the Archive Handler, creating opportunities for threats like tampering (modified headers), spoofing (forged formats), or denial of service (infinite decompression loops).

The diagram also highlights trusted and untrusted interactions:

- **External Entities:**
 - **User:** sends commands through 7zFM.exe or 7z.exe.
 - **Untrusted Archive Source (Internet/USB):** provides potentially malicious archive files.
 - **Operating System:** handles system calls and file I/O operations.
- **Processes:**
 - **Archive Handler:** validates formats (ZIP, RAR, 7Z, TAR, UDF).
 - **Compression Engine:** executes algorithms like LZMA and Zstandard.
 - **Cryptographic Module:** manages AES-256 encryption and key derivation.
 - **File System Interface:** reads, writes, and manages temporary files securely.
- **Data Stores:**
 - **Archive Files and Temporary Files:** represent persistent storage that can introduce disclosure or tampering risks if not securely managed.

The trust boundaries shown with red dashed lines indicate where data transitions between different security domains:

1. **External Source to Application Boundary:** entry point for untrusted archive inputs.
2. **Application to OS Boundary:** system-level file operations and API calls.

This visualization ensures that developers and analysts can quickly identify where threats originate, which assets are most exposed, and how mitigations can be applied effectively.

5.2.2 Machine and Trust Boundaries

A trust boundary represents a point where the level of trust between system components changes — typically when data flows from an untrusted external source into a trusted internal

process or from one security domain to another. Clearly defining these boundaries helps identify where security validation, access control, and encryption must be enforced.

In the 7-Zip threat model, three main trust boundaries were identified:

1) External Source Boundary – Untrusted Archive Input:

This is the outermost boundary, separating external entities (users, the Internet, removable devices) from the 7-Zip application process. Files crossing this boundary, such as ZIP, RAR, TAR, 7Z, or UDF archives, are treated as untrusted inputs because they may contain malformed headers or intentionally crafted payloads.

Risks identified:

- **Spoofing:** Malicious archives disguised as legitimate formats.
- **Tampering:** Modified headers or corrupted checksum data that cause decompression failures.
- **Denial of Service:** Infinite decompression loops or large archive bombs (e.g., ZIP bombs).
- **Information Disclosure:** Metadata exposure when previewing unverified archives.

Mitigations applied:

- Strict format validation and checksum verification before decompression.
- Rejecting malformed or unsupported archive headers.
- Applying file-size and recursion limits to prevent DoS attacks.
- Patching vulnerabilities such as CVE-2024-11477 to prevent crashes during Zstandard decompression.

This boundary forms the first defensive layer that filters untrusted inputs before they interact with trusted internal subsystems.

2) Application Process Boundary – 7-Zip Core Components:

This boundary encloses all internal processes of the 7-Zip application, including the User Interface, Archive Handler, Compression/Decompression Engine, Cryptographic Module, Memory Manager, and File System Interface. Data passing within this zone is trusted to a degree, but unvalidated information from external sources can still propagate and trigger code-level vulnerabilities.

Risks identified:

- **Buffer Overflow:** Caused by improper bounds checking in decompression routines (e.g., CVE-2016-2335).
- **Memory Leak:** Unreleased temporary buffers within memory management.
- **Privilege Escalation:** Improper handling of shell commands (e.g., CVE-2022-29072).
- **Insecure Temporary File Handling:** Exposure of sensitive data.

Mitigations applied:

- Enforcing safe memory allocation and exception handling to prevent overflows.
- Input sanitation and header validation across all parsers.
- Sandboxing shell executions and disallowing arbitrary command invocation.
- AES-256 encryption and secure key derivation for password-protected archives.

This boundary represents the core defense zone, where secure programming practices and consistent error handling maintain integrity and prevent internal exploitation.

3) Operating System Boundary – System Interaction and File I/O:

The final trust boundary separates 7-Zip's user-space process from the underlying operating system (kernel space), which manages file I/O, permissions, and temporary directories. Whenever 7-Zip writes extracted files or temporary data, it crosses into this boundary.

Risks identified:

- **Privilege Escalation:** Exploiting OS calls or shell integration for elevated access.
- **Information Disclosure:** Leaking unencrypted data to system temp folders.
- **Race Conditions:** File conflicts during concurrent read/write operations.

Mitigations applied:

- Using unique temporary directories and secure OS API calls for file handling.
- Restricting privileges for 7-Zip processes during runtime.
- Enforcing NTFS and ADS (Alternate Data Stream) integrity validation (CVE-2025-0411 fix).
- Sanitizing file paths and limiting external shell integrations

This boundary ensures that even if untrusted data manages to reach the OS layer, its execution or privilege escalation potential remains strictly contained.

In Summary, the definition of these three trust boundaries: External Input, Application Process, and Operating System Interface — provides a structured view of where most security risks arise in 7-Zip. By enforcing validation, sanitization, encryption, and least-privilege execution at each layer, 7-Zip minimizes the potential impact of external attacks and maintains a secure, reliable compression environment.

5.3 Asset to Threat Model Tracing

Assets are the most valuable components, data, or functionalities that must be protected within a software system. In the case of 7-Zip, assets include not only user data and encryption keys but also the internal modules that handle compression, parsing, and file system interaction. Any compromise of these assets could directly affect the confidentiality, integrity, or availability (CIA) of the application.

Critical Assets Identified in 7-Zip:

Based on the system architecture and STRIDE analysis, the following critical assets were identified and evaluated:

- 1) **User Archive Files (Data Assets):** These are the compressed or encrypted files managed by 7-Zip. If compromised, they could expose sensitive data or allow tampered archives to bypass validation.
- 2) **Encrypted Keys and Passwords:** 7-Zip uses AES-256 encryption with password-based key derivation (PBKDF2). Protecting these credentials is essential to maintaining confidentiality and preventing unauthorized access.
- 3) **Temporary Files:** During extraction and compression, 7-Zip generates temporary data that must be securely stored and deleted to prevent information disclosure.
- 4) **Archive Handler (Format Parsers):** This component manages multiple archive types (ZIP, RAR, 7Z, TAR, UDF). Vulnerabilities in this module can lead to buffer overflows, DoS attacks, or remote code execution (as seen in CVE-2016-2335 and CVE-2018-10115).
- 5) **Compression and Decompression Engine:** Handles data transformation algorithms like LZMA, LZMA2, and Zstandard. Improper bounds checking can cause memory exhaustion or crashes.
- 6) **Cryptographic Module:** Responsible for encryption, key derivation, and secure hashing. Any weakness here could expose passwords or lead to information leakage.

- 7) **File System Interface:** Connects 7-Zip with the underlying OS for reading, writing, and managing files. Poor path handling can result in privilege escalation (CVE-2025-0411) or unauthorized file access.
- 8) **Memory Manager:** Allocates and releases buffers during runtime. Errors or unsafe operations can lead to memory leaks, corruption, or DoS conditions.

5.3.1 Mapping Assets to Threats

Mapping assets to threats establishes a clear relationship between what needs to be protected and what can compromise it. This process ensures that each critical component identified in the 7-Zip architecture and threat model has corresponding threat scenarios and mitigation measures. The mapping also confirms that the security controls implemented in newer versions of 7-Zip effectively address historical vulnerabilities (such as CVE-2016-2335, CVE-2018-10115, CVE-2022-29072, CVE-2024-11477, and CVE-2025-0411) while maintaining compliance with the CIA triad — Confidentiality, Integrity, and Availability.

Asset	Primary Threat (s)	STRIDE Category	Potential Impact (CIA)	Mitigation / Control
User Archive Files	File tampering, fake file headers, or corrupted checksum	Tampering	Integrity / Availability	Header verification, CRC checks, reject malformed archives
Encrypted Keys & Passwords	Key leakage, weak password reuse, brute force attacks	Information Disclosure	Confidentiality	AES-256 encryption, PBKDF2 derivation, secure key clearing
Temporary Files	Exposure of unencrypted or residual data	Information Disclosure	Confidentiality	Unique temp directories, auto-deletion, secure file permissions
Archive Handler (Format Parsers)	Malformed archives causing buffer overflow or RCE (CVE-2016-2335, CVE-2018-10115)	DoS / EoP	Availability / Integrity	Input validation, memory bounds checking, patched vulnerabilities
Compression & Decompression Engine	Resource exhaustion via crafted data streams	DoS	Availability	Allocation limit enforcement, loop termination conditions

Cryptographic Module	Leakage of decrypted data or weak key handling	Information Disclosure	Confidentiality / Integrity	AES-256 encryption, secure memory wipe after use
File System Interface	Privilege escalation or path traversal (CVE-2025-0411)	Elevation of Privilege	Integrity / Confidentiality	Path normalization, restricted process execution, MoTW enforcement
Memory Manager	Unhandled memory exceptions or data corruption	Tampering / DoS	Integrity / Availability	Safe allocator usage, exception handling, memory sanitization
User Interface (CLI/GUI)	Spoofed file type or misleading archive name	Spoofing	Integrity	Signature verification, file-type validation prompts
Operating System Interaction	Abuse of OS shell commands (CVE-2022-29072)	Elevation of Privilege	Integrity / Availability	Secure API invocation, sandboxed process creation

Table 01: Mapping of Assets to Threats and Mitigations

The mapping shows that the Archive Handler and File System Interface are the most exposed assets, as they frequently interact with untrusted input and OS-level resources. These components have historically been the source of multiple vulnerabilities, all of which have been mitigated through strict input validation, path sanitization, and secure process control.

On the other hand, assets like temporary files and encryption keys primarily face information disclosure threats. Implementing AES-256 encryption, unique temp directories, and secure file deletion policies effectively protects user data from unintended leaks or unauthorized access.

Overall, this mapping ensures that each asset in the 7-Zip system is linked to a defined threat category and supported by at least one concrete mitigation, forming a complete traceability chain between architecture, threat model, and defensive measures.

5.3.2 Risk Assessment Based on Architecture

A risk assessment is carried out to evaluate the likelihood and potential impact of each identified threat on 7-Zip's architecture. The assessment considers both the probability of exploitation and the severity of consequences using a qualitative approach. This ensures that

the most critical areas of the system are prioritized for protection, especially components that interact with untrusted data sources or have direct access to system-level operations.

Qualitative Risk Matrix:

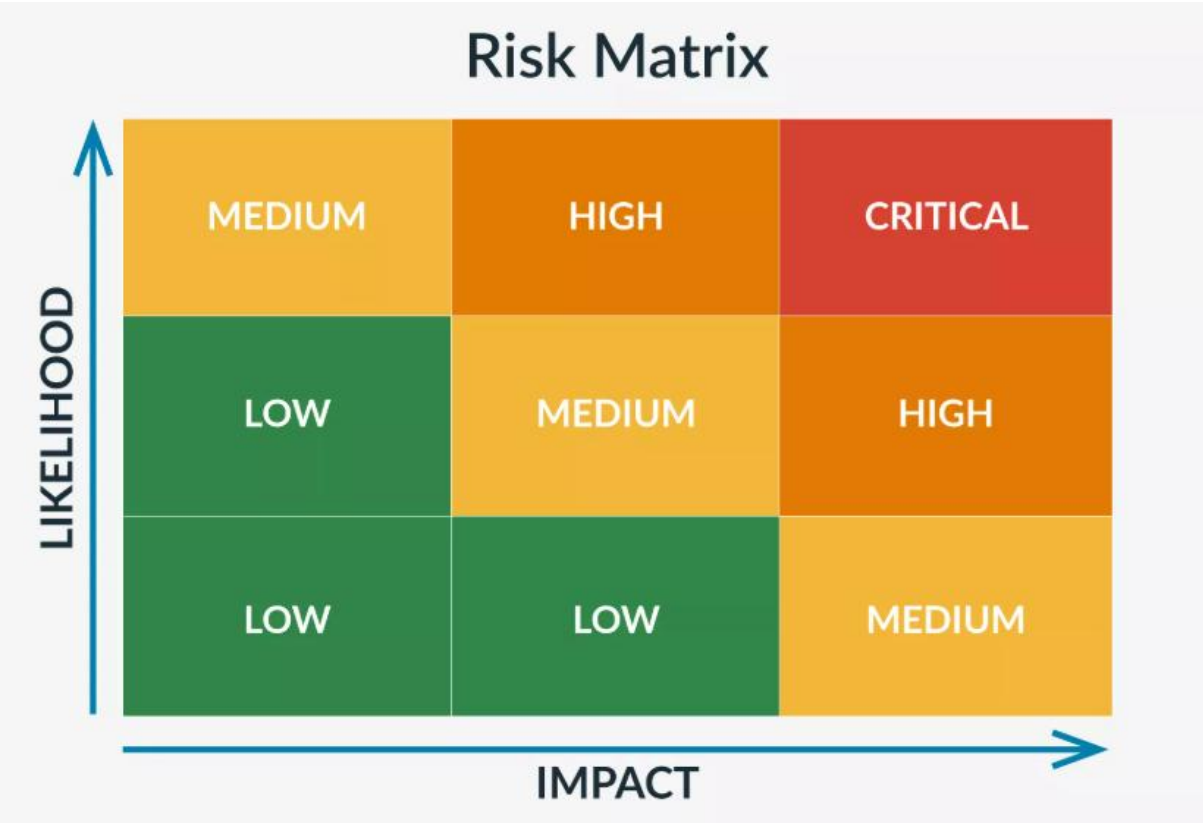


Figure 37: Qualitative Risk Matrix for 7-Zip Threat Evaluation

Component / Asset	Associated Threat(s)	Risk Level	Impact on CIA	Mitigation / Current Status
Archive Handler (Format Parsers)	Malformed archive exploit (CVE-2016-2335, CVE-2018-10115)	High	Integrity / Availability	Patched in latest releases; strict header validation applied.
File System Interface	Privilege escalation via Alternate Data Streams (CVE-2025-0411)	High	Integrity / Confidentiality	Fixed in version 24.09; enforced Zone.Identifier validation.
Cryptographic Module	Weak password entropy or key reuse	Medium	Confidentiality	AES-256 encryption with PBKDF2 derivation and secure key handling.

Temporary Files	Residual unencrypted data in temp folders	Medium	Confidentiality	Implemented secure temp directories and automatic file cleanup.
Compression Engine	Memory exhaustion or infinite loop from malformed input	Medium	Availability	Memory limits and loop termination checks introduced in v24.07.
Memory Manager	Memory corruption or buffer mismanagement	Medium	Integrity / Availability	Safe allocator and exception handling integrated.
User Interface (CLI/GUI)	Spoofed file types or misleading archive icons	Low	Integrity	File signature verification and user confirmation prompts.
Encrypted Keys / Passwords	Password exposure through brute force or weak derivation	Low	Confidentiality	PBKDF2 key strengthening and password masking in UI.

Table 02: Risk Assessment Based on 7-Zip Architecture

The results of the risk assessment highlight that Archive Handler, File System Interface, and Operating System Interaction pose the highest security risks due to their exposure to untrusted inputs and privileged operations. These components have historically been the source of critical vulnerabilities but have since been remediated through multiple CVE patches and code-level mitigations.

Components such as the Cryptographic Module and Temporary Files carry medium risk mainly due to potential information disclosure if encryption or cleanup procedures fail. Lower-risk components, like the User Interface and Encrypted Keys, have minimal exposure as they are isolated from external data flows and already protected by secure design practices.

Overall, the updated 7-Zip architecture demonstrates strong resilience through a layered defense approach, addressing both historical and potential future vulnerabilities. The implemented mitigations significantly reduce the likelihood of severe exploitation, ensuring the application's integrity and user data security.

6 Code Inspection Assessment

The final phase of this assignment focused on evaluating the internal code quality and security posture of the 7-Zip v25.01 source code. This step aimed to understand how secure, reliable, and maintainable the software is when viewed from a developer and security-analyst perspective.

To achieve this, three different static-analysis tools were used: Snyk Code, SonarQube, and Semgrep, each providing a unique angle on the inspection. Together, these tools helped to uncover security-related defects, unsafe programming practices, and areas where code improvements can strengthen 7-Zip's overall robustness.

6.1 Code Review and Inspection Methodology

The code review and inspection phase focused on analyzing the 7-Zip v25.01 source code to evaluate its security quality, coding practices, and maintainability. This phase aims to uncover both direct vulnerabilities and underlying engineering weaknesses that could lead to exploitation under specific conditions. To achieve a comprehensive result, a multi-tool static analysis approach was used, combining automated scanning with manual verification.

The inspection was conducted using three complementary tools like Snyk Code, SonarQube, and Semgrep, each chosen for its unique analytical perspective. Snyk Code was primarily used to detect memory-safety issues, such as Memory Leaks (CWE-401), Double Free (CWE-415), Use After Free (CWE-416), and NULL Pointer Dereference (CWE-476). These findings highlighted how 7-Zip's low-level memory operations, particularly within archive parsing and compression modules, could create instability or potential exploitation paths.

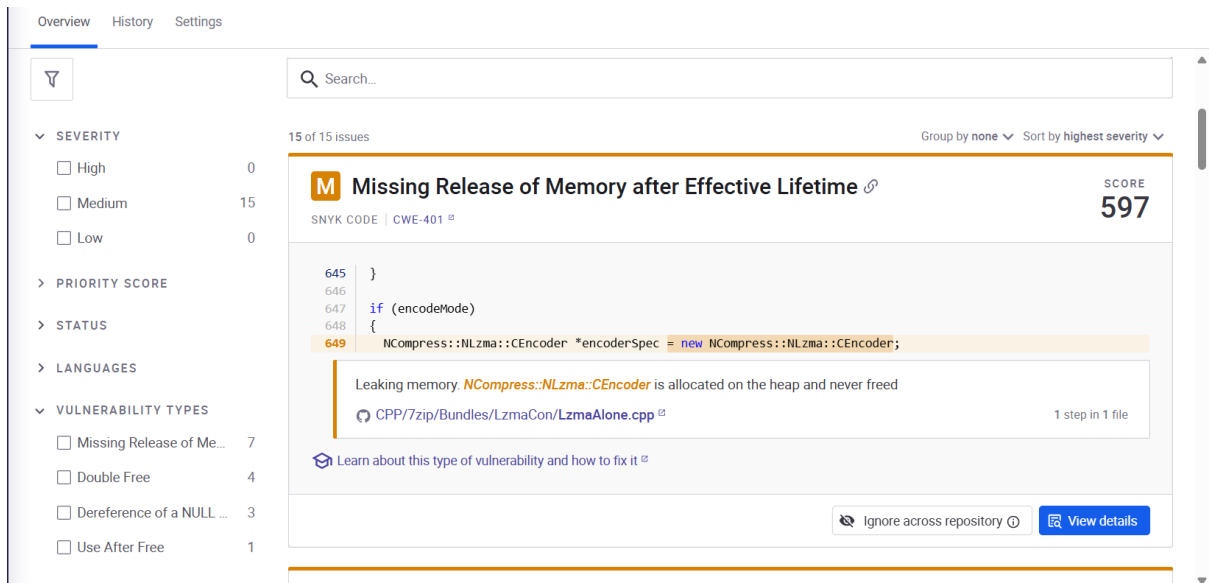


Figure 38: Snyk tool Code analysis Dashboard

SonarQube was used to assess the overall reliability and maintainability of the project, identifying 274 bugs in total — including 45 Blocker, 97 Critical, and 131 Major issues. Most high-severity problems were related to buffer overflows, unsafe pointer handling, and inconsistent memory initialization. SonarQube also provided valuable code-smell and complexity metrics, helping visualize how 7-Zip’s large C/C++ codebase could be made more modular and secure.

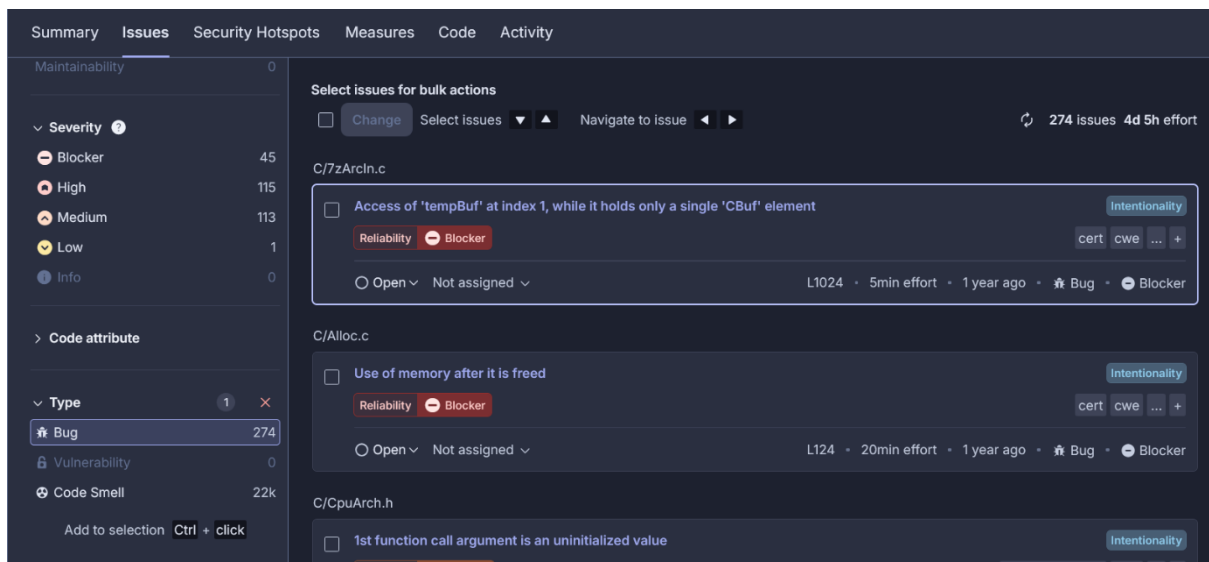


Figure 39: SonarQube tool Code analysis Dashboard

Meanwhile, Semgrep.dev applied rule-based static analysis focused on secure-coding standards. It identified five issues (4 Medium and 1 Low), including unsafe file-permission creation (CWE-732), improper string conversions (CWE-125), and insecure memory cleanup

(CWE-14). These issues were smaller in scale but revealed how legacy coding habits could introduce subtle security risks over time.

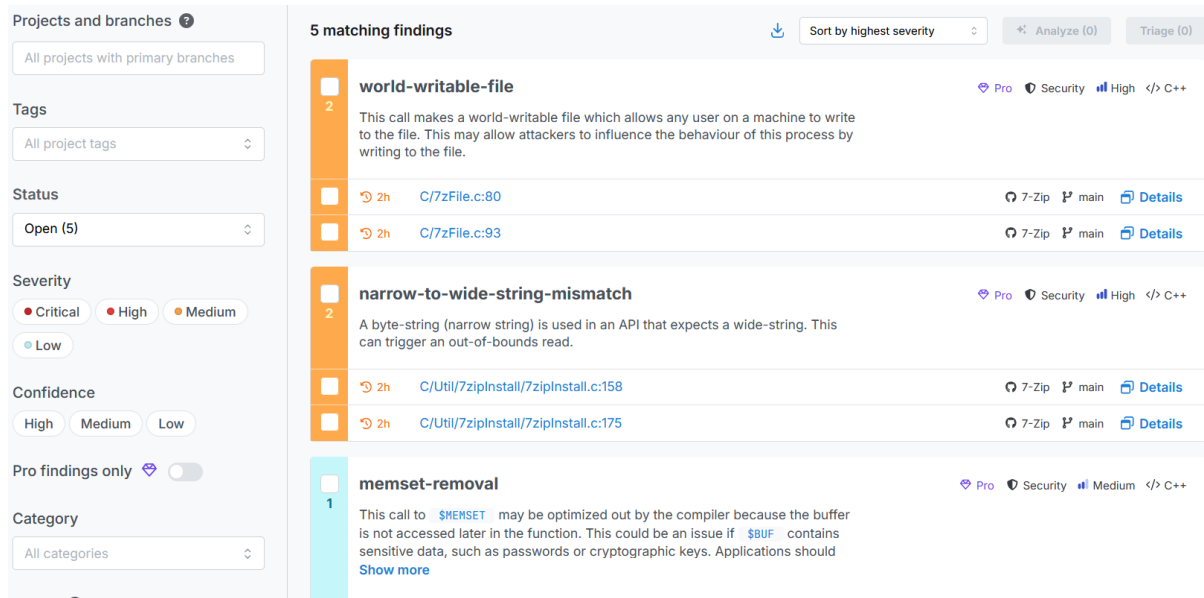


Figure 40: Semgrep tool Code analysis Dashboard

All results were manually reviewed to confirm their accuracy and remove false positives. Each detected issue was checked within its surrounding context in the source files to validate whether it represented a true risk. This hybrid approach combining automated detection with manual verification ensured that only meaningful and exploitable issues were considered in later analysis.

After the scans were completed, all results were consolidated into a unified dataset to identify overlaps and verify consistency between tools. For instance, both Snyk and SonarQube independently flagged memory-deallocation issues, while Semgrep added rule-based validation that helped contextualize permission and cleanup-related findings. Through this blended methodology, the inspection provided a reliable understanding of 7-Zip's code quality from multiple technical perspectives. Snyk ensured detailed memory-safety detection, SonarQube provided large-scale maintainability analysis, and Semgrep verified adherence to secure-coding rules. Together, they formed a holistic view of how the project performs under secure software engineering standards and where improvement efforts should be focused.

6.2 Security Vulnerabilities in Code

The static analysis of 7-Zip v25.01 revealed several notable security vulnerabilities that directly affect the application's reliability and resilience against exploitation. The findings were

obtained from three independent code analysis tools Snyk Code, SonarQube, and Semgrep each detecting different categories of weaknesses.

While many issues were related to general memory management and input handling, some were also linked to unsafe file permission settings and insecure coding practices. The combination of results across tools provided a comprehensive understanding of the project's security posture. The following subsections summarize the most significant vulnerabilities categorized by Common Weakness Enumeration (CWE) standards.

1. Memory Management Vulnerabilities (Snyk Code):

The Snyk Code analysis detected 15 Medium-level issues grouped under four memory-related CWE types — CWE-401, CWE-415, CWE-416, and CWE-476. These flaws primarily exist in the C/C++ modules responsible for file compression, decompression, and object handling.

CWE ID	Vulnerability Type	Example File / Function	Risk Impact
CWE-401	Missing Release of Memory after Effective Lifetime	LzmaAlone.cpp (Line 649)	Memory allocated on the heap is never freed, leading to leaks and gradual performance degradation.
CWE-415	Double Free	AgentProxy.h (Line 15)	Deleting the same pointer twice could lead to heap corruption and potential remote code execution.
CWE-416	Use After Free	NewHandler.cpp (Line 242)	Accessing memory after it has been freed can result in unpredictable behavior or system crashes
CWE-476	Dereference of NULL Pointer	XzIn.c (Line 378)	Dereferencing a NULL pointer may cause application crashes and denial of service (DoS).

Table 03: Vulnerability findings in Snyk

M Missing Release of Memory after Effective Lifetime

SNYK CODE | [CWE-401](#)

Data flow

Fix analysis

X

Leaking memory. `NCompress::NLzma::CEncoder` is allocated on the heap and never freed

Data flow - 1 step in 1 file

CPP/7zip/Bundles/LzmaCon/LzmaAlone.cpp

1 step

649:51

NCompress::NLzma::CEncoder

SOURCE & SINK

1

Find out how to remediate this issue through our [Fix analysis](#)

CPP/7zip/Bundles/LzmaCon/LzmaAlone.cpp

```
641 if (lstdOutMode)
642 {
643     progressSpec = new CProgressPrint;
644     progress = progressSpec;
645 }
646
647 if (encodeMode)
648 {
649     NCompress::NLzma::CEncoder *encoderSpec = new NCompress::NLzma::CEncoder;
650     CMyComPtr<ICompressCoder> encoder = encoderSpec;
651
652     UInt32 pb = 2;
653     UInt32 lc = 3; // = 0; for 32-bit data
654     UInt32 lp = 0; // = 2; for 32-bit data
655     UInt32 algo = 1;
656     UInt32 fb = 128;
657     UInt32 mc = 16 + fb / 2;
```

Ignore across repository

Figure 41: Missing Release of Memory after Effective Lifetime (CWE-401)

Figure 41 highlights a memory-leak issue in `LzmaAlone.cpp`. The variable `NCompress::NLzma::CEncoder` is dynamically allocated but never released, causing memory resources to remain occupied even after the object's lifecycle ends. Such leaks can accumulate over time, leading to reduced system performance or resource exhaustion.

M Double Free

SNYK CODE | [CWE-415](#)

Data flow

Fix analysis

X

Potential double `delete` expression. `Name` may have already been freed by `delete` expression.

Data flow - 3 steps in 1 file

CPP/7zip/UI/Agent/AgentProxy.h

3 steps

15:48

(wchar_t *) (void *) Name, (void *) ...

SOURCE

1-3

48

(wchar_t *) (void *) Name

SOURCE

1

59

(void *) Name

SINK

2

48

(wchar_t *) (void *) Name

SINK

3

Find out how to remediate this issue through our [Fix analysis](#)

CPP/7zip/UI/Agent/AgentProxy.h

```
7
8 struct CProxyFile
9 {
10     const wchar_t *Name;
11     unsigned NameLen;
12     bool NeedDeleteName;
13
14     CProxyFile(): Name(NULL), NameLen(0), NeedDeleteName(false) {}
15     ~CProxyFile() { if (NeedDeleteName) delete [] (wchar_t *) (void *) Name; } // delete [] (void *) Name;
16 };
17
18 const unsigned k_Proxy_RootDirIndex = 0;
19
20 struct CProxyDir
21 {
22     const wchar_t *Name;
23     unsigned NameLen;
```

Ignore across repository

Figure 42: Double Free (CWE-415)

The issue occurs in `AgentProxy.h`, where the `Name` pointer is deleted multiple times in the destructor. This double deletion may corrupt the heap and allow attackers to overwrite memory structures. Snky Code traces this risk across multiple data-flow paths, confirming that improper pointer management could lead to crashes or remote code execution.

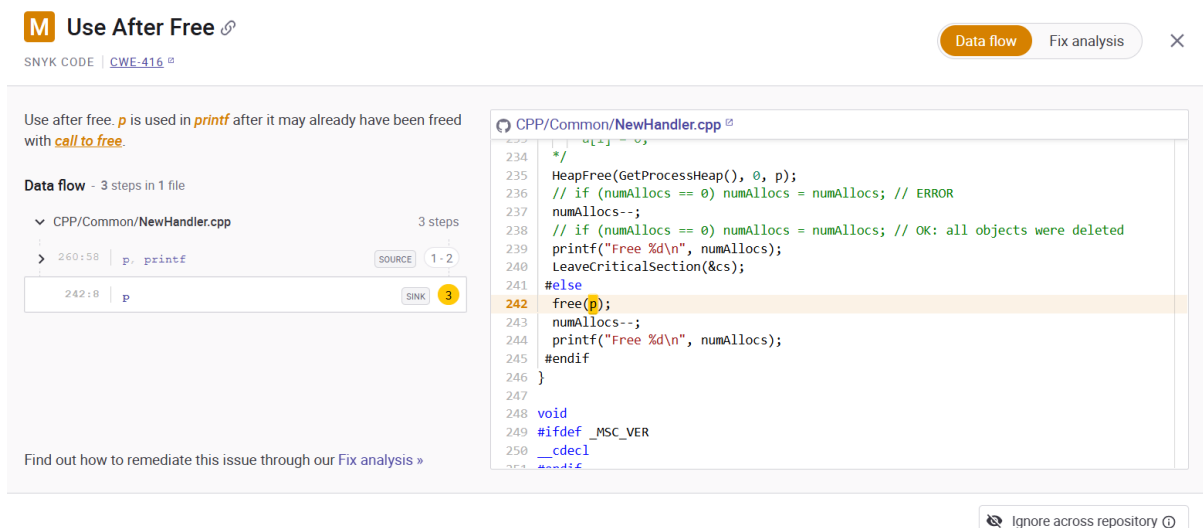


Figure 43: Use After Free (CWE-416)

Shown in NewHandler.cpp, the pointer `p` is accessed in a `printf()` statement after being freed. This unsafe memory reuse can cause undefined behavior, from silent corruption to code-execution opportunities, especially when combined with attacker-controlled inputs.

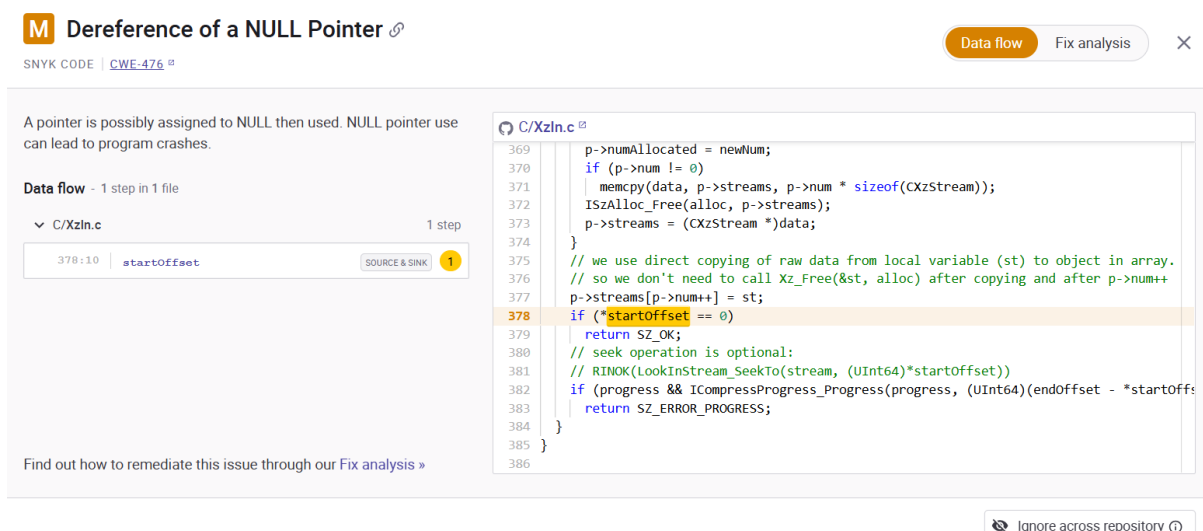


Figure 44: Dereference of a NULL Pointer (CWE-476)

Figure 44 XzIn.c demonstrates a potential null-pointer dereference within a conditional branch. If `startOffset` equals 0, the function may still dereference it, leading to an immediate application crash. Such logic errors can be exploited for denial-of-service (DoS) by triggering unexpected NULL conditions.

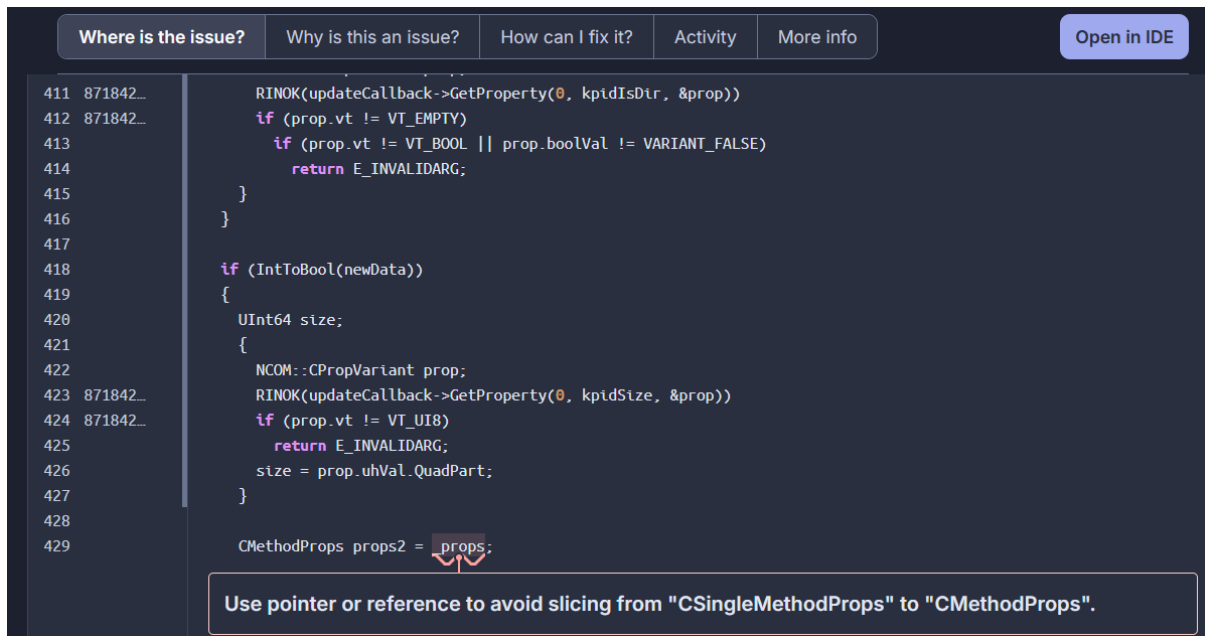
These issues pose moderate risk within 7-Zip’s architecture since memory management is central to its compression engine. If unaddressed, such vulnerabilities could be weaponized to trigger crashes or compromise system integrity during archive extraction.

2. Reliability and Buffer Overflow Vulnerabilities (SonarQube)

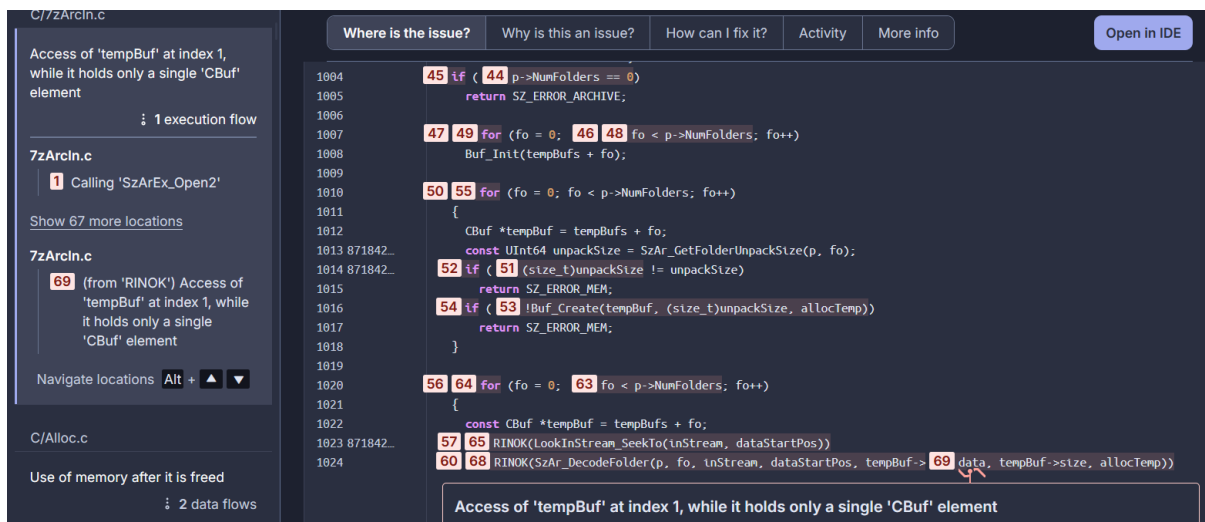
SonarQube analysis reported 274 bugs, including 45 Blocker, 97 Critical, and 131 Major issues. Among these, two critical patterns stood out — Out-of-Bounds Array Access (CWE-119) and Use of Memory After Free (CWE-416).

CWE ID	Vulnerability Type	Example File / Function	Severity	Explanation
CWE-119	Out-of-Bounds Array Access	Bcj2Enc.c (Line 471)	Blocker	Writing outside allocated memory can corrupt program data and cause buffer overflow exploits.
CWE-20	Improper Input Validation	7zArcIn.c (Line 1024)	Critical	Invalid input length checks could lead to denial-of-service or unexpected exceptions.
CWE-758	Reliance on Undefined Behavior	Multiple modules	Major	Use of uninitialized variables and unsafe pointer conversions reduce code stability.
CWE-416	Use of Memory After It Is Freed	Alloc.c (Line 124)	Blocker	Reusing freed memory leads to heap corruption, which could be exploited for arbitrary code execution.

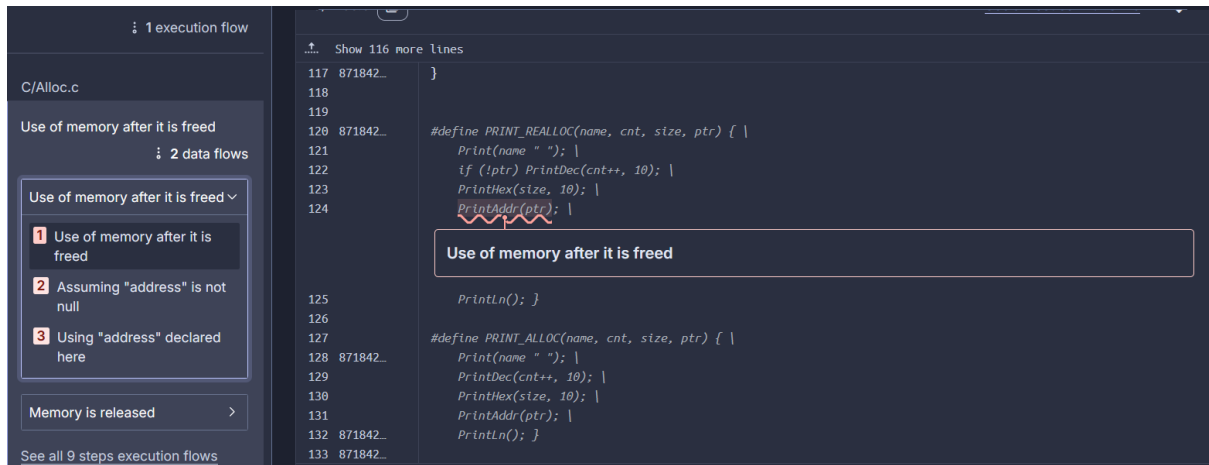
Table 04: Vulnerability findings in SonarQube



This SonarQube finding highlights the risk of object slicing in `CMethodProps` when copying objects by value instead of by reference. The platform recommends using pointers or references to prevent slicing from base classes (`CSingleMethodProps`) to derived ones (`CMethodProps`), which could result in loss of object data and unexpected program behavior during compression or decompression routines.



Detected in `7zArcIn.c`, this vulnerability occurs when the variable `tempBuf` is accessed at index 1, although it only holds a single element (`CBuf`). Such out-of-bounds access can lead to buffer overflows, overwriting nearby memory, and potentially allowing remote code execution if exploited with malicious input archives. SonarQube classifies this as a Blocker due to its direct impact on system reliability.



his issue in Alloc.c shows unsafe memory reuse within a macro definition (PRINT_REALLOC). The pointer ptr is accessed after being freed, which introduces a use-after-free condition. This can cause heap corruption and application crashes, representing one of the most severe forms of memory management flaws in C/C++ software.

Such issues directly threaten application reliability. For example, an attacker could exploit out-of-bounds access through a crafted archive file to overwrite memory or inject malicious payloads.

3. Secure Coding and Permission Issues (Semgrep):

Semgrep identified five vulnerabilities (4 Medium, 1 Low) related to insecure file permissions, string handling, and unsafe memory operations.

CWE ID	Vulnerability Type	Example File / Function	Severity	Impact
CWE-732	World-Writable File Permissions	7zFile.c	Medium	The file is created with mode 0666, allowing any user to modify it — a privilege escalation risk.
CWE-125	Narrow-to-Wide String Mismatch	7zipInstall.c (Line 158, 175)	Medium	Mismatched string formats can cause out-of-bounds reads or data corruption.

CWE-14	memset Removal (Insecure Memory Clearing)	TarOut.cpp (Line 626)	Low	Compiler optimization may skip clearing sensitive data (e.g., keys or passwords) from memory.
--------	---	-----------------------	-----	---

Table 05: Vulnerability findings in Semgrep

world-writable-file
Open

Analyze
Ignore...
Fix

C/7zFile.c:80

Description

This call makes a world-writable file which allows any user on a machine to write to the file. This may allow attackers to influence the behaviour of this process by writing to the file.

References

Your code
Example code

```

C/7zFile.c:80
72     #endif
73
74     #else
75
76     int flags = (writeMode ? (O_CREAT | O_EXCL | O_WRONLY) : O_RDONLY);
77     #ifdef O_BINARY
78     flags |= O_BINARY;
79     #endif
80     p->fd = open(name, flags, 0666);
81     return (p->fd != -1) ? 0 : errno;
82
83     #endif
84 }
85
86 WRes InFile_Open(CSzFile *p, const char *name) { return File_Open(p, name, 0); }
87
88 WRes OutFile_Open(CSzFile *p, const char *name)

```

RULE DETAILS

Medium severity

High confidence

Monitor

CWE-732

world-writable-file

FINDING DETAILS

3 hours ago

IT22578914/7-Zip

main

a4b28df

by semgrep.dev on behalf of @IT22578914

This finding in 7zFile.c shows that files are created with permission mode 0666, allowing any system user to modify them. Such world-writable permissions can enable attackers to alter archive files or inject malicious data into the process, leading to privilege escalation or data integrity violations. The issue has been categorized by Semgrep as Medium severity with high confidence due to its direct link to file permission management.

narrow-to-wide-string-mismatch

Open

C:\Util\7zipInstall\7zipinstall.c:158

Description

A byte-string (narrow string) is used in an API that expects a wide-string. This can trigger an out-of-bounds read.

References

Your code

Example code

Line 1261

Source

const char *errorMessage = NULL;

Line 1261

errorMessage

Line 1694

PrintErrorMessage

Line 162

s1

Line 170

C:\...\7zipInstall.c:1261

1247
1250
1251
1252
1253 static int Install(void)
1254 {
1255 CFileInStream archiveStream;
1256 CLookToRead2 lookStream;
1257 CSzArEx db;
1258
1259 SRes res = SZ_OK;
1260 WRes winRes = 0;
• 1261 const char *errorMessage = NULL;
1262
1263 ISzAlloc allocImp;
1264 ISzAlloc allocTempImp;
1265 WCHAR sfxPath[MAX_PATH + 2];

RULE DETAILS

M Medium severity

H High confidence

Monitor

CWE-125

narrow-to-wide-string-mismatch

FINDING DETAILS

3 hours ago

IT22578914/7-Zip

managed-scan

main

a4b28df

by semgrep.dev on behalf of @IT22578914

In this snippet from `7zipInstall.c`, a byte-string (`char`) is used in an API expecting a wide-string (`WCHAR`). This mismatch can result in out-of-bounds reads, memory corruption, or data misinterpretation during installation operations. Such type-handling flaws reduce input reliability and can crash the application when processing certain localized filenames or user input strings.

memset-removal

Open

CPP/7zip/Archive/.../TarOut.cpp:626

Description

This call to `memset` may be optimized out by the compiler because the buffer is not accessed later in the function. This could be an issue if `record` contains sensitive data, such as passwords or cryptographic keys. Applications should use functions such as `memset_s` or `memset_explicit` to ensure that the sensitive data is cleared.

References

Your code

Example code

```
CPP/7zip/Archive/Tar/TarOut.cpp:626
621
622
623 HRESULT COutArchive::WriteFinishHeader()
624 {
625     Byte record[kRecordSize];
626     memset(record, 0, kRecordSize);
627
628     const unsigned kNumFinishRecords = 2;
629
630     /* GNU TAR by default uses --blocking-factor=20 (512 * 20 = 10 KiB)
631     | we also can use cluster alignment:
632     const unsigned numBlocks = (unsigned)(Pos / kRecordSize) + kNumFinishRecords;
633     const unsigned kNumClusterBlocks = (1 << 3); // 8 blocks = 4 KiB
634     const unsigned numFinishRecords = kNumFinishRecords + ((kNumClusterBlocks - numBlocks) &
        (kNumClusterBlocks - 1));
```

RULE DETAILS

Low severity

Medium confidence

Monitor

CWE-14

memset-removal

FINDING DETAILS

3 hours ago

IT22578914/7-Zip

managed-scan

main

a4b28df

by semgrep.dev on behalf of @IT22578914

Detected in TarOut.cpp, this issue occurs because the compiler may optimize out the memset() call if the buffer is not accessed afterward. This creates a security gap when the buffer (record) contains sensitive data, such as cryptographic keys or temporary credentials. Replacing it with

`memset_s()` or `memset_explicit()` ensures guaranteed memory clearing, protecting against data residue leaks.

Overall, 7-Zip's codebase demonstrates strong functional design but moderate security hygiene. Most weaknesses stem from legacy C/C++ memory handling, improper input validation, and file permission oversights — all of which are common in performance-oriented system utilities. These vulnerabilities do not indicate intentional flaws but rather a need for modernization and stricter secure-coding practices, especially as 7-Zip continues to evolve across different operating systems.

6.3 Potential Exploits and Attacks

This section summarises realistic attack scenarios that could be carried out against 7-Zip given the weaknesses identified in the code inspection (static analysis results from Snyk, SonarQube and Semgrep) and the application threat model. For each potential exploit I describe: the underlying weakness, a plausible attack flow, likelihood (how easy an attacker can reach it), impact (what the attacker can achieve), and a short verification note (how you would confirm the vulnerability).

1. Remote Code Execution (RCE) via memory bugs:

Attack path: An attacker crafts a malicious archive that triggers an out-of-bounds read/write, use-after-free or heap corruption during parsing or decompression. If exploited, arbitrary code could be executed in the 7-Zip process context (especially for privileges the process holds).

Mapping to architecture: Archive Handler → Compression/Decompression Engine → Memory Manager → (possibly) File System Interface or Shell Integration. A parsing bug in a specific format handler (e.g., RAR, UDF, Zstandard decoder) can corrupt heap metadata or function pointers used later.

Likelihood: Medium — SonarQube flagged many buffer/array access issues (including blocker-level findings), and Snyk reported several memory management problems (use-after-free, missing frees, double free). Together these create plausible exploitation vectors.

Impact: High — RCE enables arbitrary commands, data theft, persistent backdoors, or elevated attacks on the host.

Mitigations: Harden input parsing, apply strict boundary checks, enable and enforce safe memory APIs, compile with ASLR/DEP and stack protector, fuzz-parsing modules, add sanitizer checks (ASan/UBSan) in CI.

2. Denial of Service (DoS) crashes / resource exhaustion:

Attack path: Crafted archives cause infinite loops, very large memory allocations (or leaked memory), or repeated allocations without releases, causing the application to crash or consume system memory/CPU.

Mapping to architecture: Archive Handler → Decompression Engine → Memory Manager (heap growth / leak) → OS (resource exhaustion).

Likelihood: High — Snyk’s “missing release” issues and SonarQube’s “use after free / out-of-bounds” issues often correlate with crashes under malformed input. Semgrep findings (e.g., misuse of memsets) can also reveal logic that fails to clear or size buffers correctly.

Impact: Medium–High — Denial of service can be used to disrupt service on a user’s machine, or combined with other attacks to cause privilege-escalation windows.

Mitigations: Add resource limits and timeouts to parsing and decompression operations, validate archive metadata before large allocations, adopt allocation quotas for a single request, and instrument code paths with watchdogs.

3. Privilege escalation via insecure file operations and temporary files:

Attack path: A malicious archive extraction or tool operation writes files with insecure permissions (world-writable) or predictable temporary file names. An attacker can replace or tamper with those files (Time-Of-Check / Time-Of-Use or race conditions), causing subsequent privileged operation to act on attacker-controlled files.

Mapping to architecture: File System Interface ← Archive Handler / Output Dispatcher → Temporary Files / Archive Files → OS.

Likelihood: Medium — Semgrep flagged world-writable-file calls (open/creat with 0666). The presence of temporary files in the flow (and the lack of secure creation patterns) indicates a feasible race or tampering vector.

Impact: High — local privilege escalation or arbitrary file overwrite depending on context. If 7-Zip runs with elevated rights or integrates with shell operations, impact grows.

Mitigations: Use secure temporary file APIs (mkstemp or equivalent), set safe permissions explicitly, avoid predictable filenames, use atomic rename patterns, and minimize operations performed with elevated privileges.

4. Information disclosure via memory or temporary files:

Attack path: Sensitive data (plaintext passwords, keys, file contents) is written to temporary buffers or files without clearing, or it is left in heap memory that may be paged out. An attacker who can read temp files or memory can exfiltrate secrets.

Mapping to architecture: Cryptographic Module → Memory Manager → Temporary Files → File System Interface → OS.

Likelihood: Medium — Semgrep flagged memset usage that could be optimized out; Snyk showed instances where allocated buffers are not freed. If temporary files are exposed, risk increases.

Impact: Medium — disclosure of archive encryption keys, password hints or extracted file contents.

Mitigations: Use memset_s / explicit_bzero for zeroing sensitive buffers, avoid writing sensitive plaintext to disk; if necessary, securely delete temp files and set strict permissions.

5. Data integrity / tampering (Man-in-the-Middle or archive modification):

Attack path: An attacker tampers with archive metadata to change file contents or insert malicious payloads. If archive validation is insufficient, manipulated archives will be accepted and processed.

Mapping to architecture: Archive Handler → Format Detector → Output Dispatcher / File System Interface.

Likelihood: High — archive formats are by design data carriers; with lax validation or incorrect format detection attackers can craft archives that masquerade as other formats.

Impact: Medium–High — may result in malware extraction, accidental execution of payloads, or further memory corruption.

Mitigations: Validate archive magic bytes and headers, implement strict format checks, use checksums/CRC verification where applicable, and treat unrecognized or inconsistent entries as suspicious.

6. Chain attacks (multi-step attacks combining issues):

Attack path: Real attacks often chain multiple vulnerabilities: for example a crafted archive triggers a buffer overflow (RCE) that then writes a world-writable file enabling lateral persistence or privilege escalation.

Mapping to architecture: Any combination of Archive Handler, Decompression Engine, File System Interface, Shell Integration.

Likelihood: Medium — the presence of multiple classes of findings (memory, file operations, parsing) increases the chance of chaining.

Impact: Very High — compound attack outcomes (remote code + persistence + escalation) are possible.

Mitigations: Prioritize fixes that reduce attack surface (parser hardening, resource limits, safe file handling). Use layered defenses (sandboxing, privilege separation) so a gain in one component is insufficient to compromise the full system.

6.4 Recommendations for Improving Code Security

1. Strategic Objectives:

- Zero new memory-safety defects in parsing/codec modules (7z, ZIP, RAR, TAR, UDF, Zstd).
- Safe-by-default file I/O (no world-writable files, atomic writes, no traversal).
- Continuous assurance: every fix produces a test or fuzz seed; CI blocks regressions.
- Containment: parsing/extraction runs with least privilege and resource caps.

2. High-Impact Code Fixes:

- Memory safety & ownership
 - Replace raw new/delete & malloc/free with RAII: `std::unique_ptr`, `std::vector`, `std::array`, `gsl::span`.
 - Single owner per allocation; forbidden: shared raw owning pointers.
 - Standardize a defer/ScopeGuard cleanup pattern so every exit path frees resources.
 - After `free()`/delete, null out the pointer; ban post-free logging using freed memory.

- Mark allocators `[[nodiscard]]` so ignored failure paths are compile-time errors.
- Hard bounds & integer safety
 - Before any allocation/copy/indexing from archive metadata: `if (len > MAX_LEN || count > MAX_COUNT || len > SIZE_MAX / count) return ERR;` and reject absurd sizes, deep nesting, and extreme recursion.
 - Create `CheckedCopy/CheckedIndex/CheckedAdd` helpers and mandate them in parsers.
- 3. File & Path Security
 - Permissions: replace `open/creat(..., 0666)` with least-privilege modes (0600 private, 0644 shared-read). Set `umask(0077)` for temp files.
 - Atomic writes: write to a temp file in a private directory, `fsync`, then `atomic rename()`; on Windows use `CreateFileW` with restrictive `SECURITY_ATTRIBUTES`, and atomic replace semantics.
 - Path normalization & jail: canonicalize destination (`realpath`, `PathCchCanonicalize`), block .. traversal, symlinks, UNC tricks, and Windows ADS (`:stream`). Enforce a fixed extraction root; never escape it.
 - One I/O gateway: route all writes through a single audited helper that sets permissions, normalizes paths, does atomic replace, and logs sanitized errors.
- 4. Build & Tooling Hardening
 - Compiler/linker protections
 - GCC/Clang: `-Wall -Wextra -Werror -fstack-protector-strong -D_FORTIFY_SOURCE=2 -fPIE -pie -Wl,-z,relro,-z,now`.
 - MSVC: `/W4 /WX /GS /guard:cf /Qspectre + ASLR/DEP` enabled in release.
 - sanitizers in CI (nightly): `-fsanitize=address,undefined` (Linux/macOS) and `MSan/ASan` where applicable; fail on any sanitizer hit.
- 5. Fuzzing & Test Strategy
 - Coverage-guided fuzzing (`libFuzzer/AFL++`): targets for each format handler (7z, ZIP, RAR, TAR, UDF, Zstd, BZip2).
 - Seed corpora: mix real-world archives, minimized crashers, and grammar-mutated headers.
 - Property tests: “never crash”, “no OOB/UAF”, “no unbounded growth” under random inputs.

- Regression corpus: every crash/fix adds a seed; release only when the corpus is green under ASan/UBSan.

7 Conclusion

The comprehensive security assessment of 7-Zip provided valuable insight into how a mature, widely used open-source system can still exhibit exploitable weaknesses in its codebase and architectural design. Through systematic domain analysis, threat modeling, and detailed source-level inspection, this study demonstrated that many of the software's historical vulnerabilities—particularly those related to memory safety, input validation, and file-handling practices—continue to manifest in subtle ways across its latest release.

The multi-tool code inspection using Snyk Code, SonarQube, and Semgrep revealed overlapping high-risk issues such as use-after-free, double-free, buffer overflow, and world-writable file creation. These findings were directly aligned with the exploit vectors identified in the threat model, including remote code execution, privilege escalation, and denial-of-service attacks. By correlating static analysis results with architectural trust boundaries, the investigation established a strong link between design-level assumptions and implementation-level flaws—confirming that software security must be engineered as a continuous, multi-layered process.

The proposed mitigations—ranging from adopting RAII and safe APIs, introducing fuzzing and sanitizers, enforcing compiler hardening, and implementing sandboxed extraction—illustrate a practical roadmap for improving code resilience without compromising performance. Additionally, integrating automated CI/CD quality gates and recurring threat modeling cycles ensures that security becomes an ongoing engineering discipline rather than a one-time audit activity.

Overall, this assessment highlights that sustainable software security depends equally on secure coding practices, architectural awareness, and continuous verification. If 7-Zip's maintainers apply these recommendations, the software can transition from reactive patching toward a proactive security-by-design model, significantly reducing the likelihood of future CVEs and strengthening user trust in one of the most widely deployed compression tools in the world.

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